ÖAW AUSTRIAN ACADEMY OF SCIENCES

austrian centre for digital humanities

My first DH Toolbox

CBS4DH 2026W, 22 October 2026

Massimiliano Carloni, Dimitra Grigoriou,
Kevin Stadler, Katharina Wünsche

With material also contributed by
Nina C. Rastinger

General notes on tool collection

- selection of tools useful for DH (but not only)
- online/web & offline/desktop applications
- (mainly) open & proprietary software
- ‘ever-growing’ and open collection (non-exhaustive and exemplary selection)
- different tools – different purposes

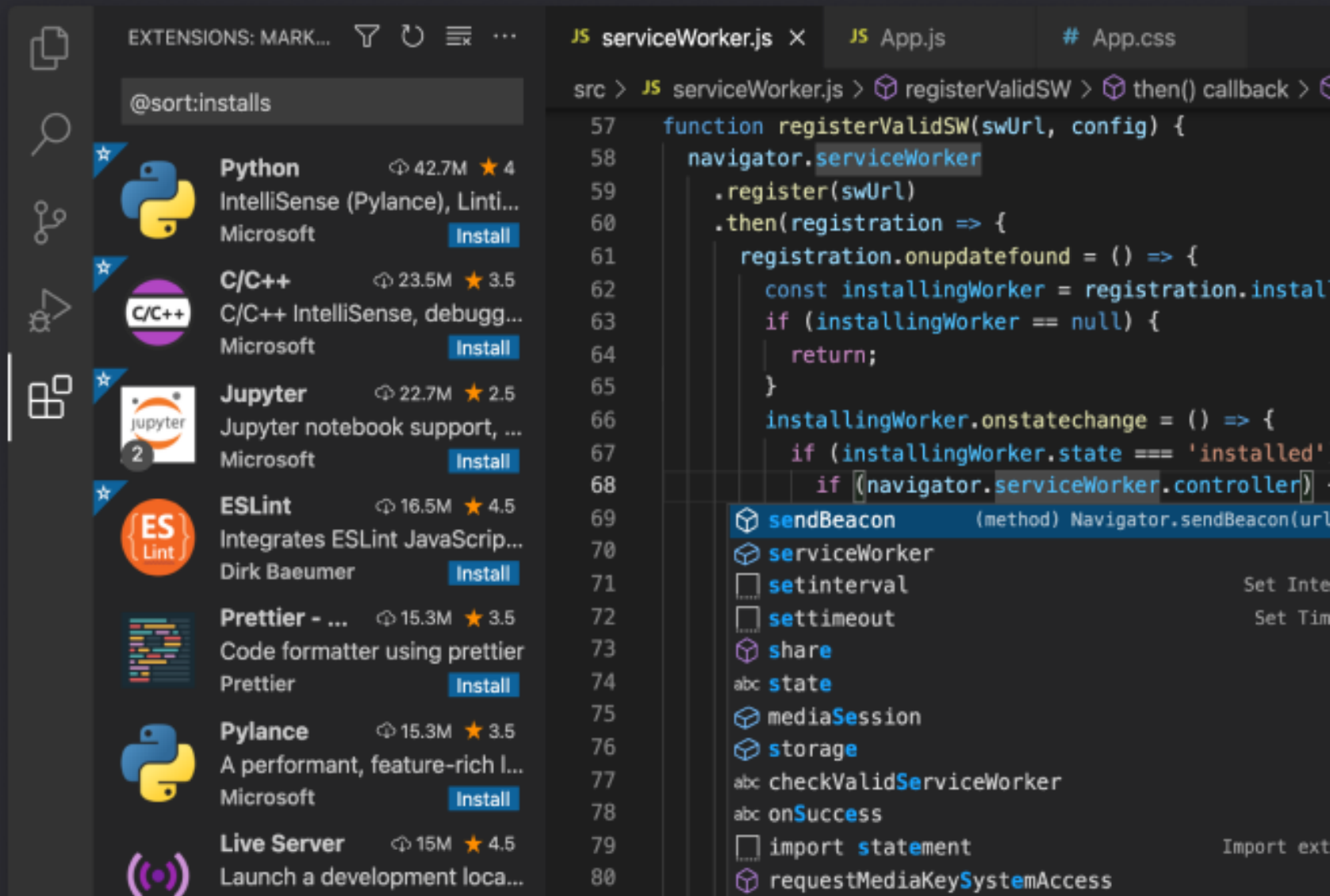
Tool collection

- Code editors and IDEs
- Automated Text Recognition
- Annotation
- Data Collection & Organisation
- Data Analysis & Visualisation
- AI, GIS, 3D Visualisation, and more...

Code editors and IDEs

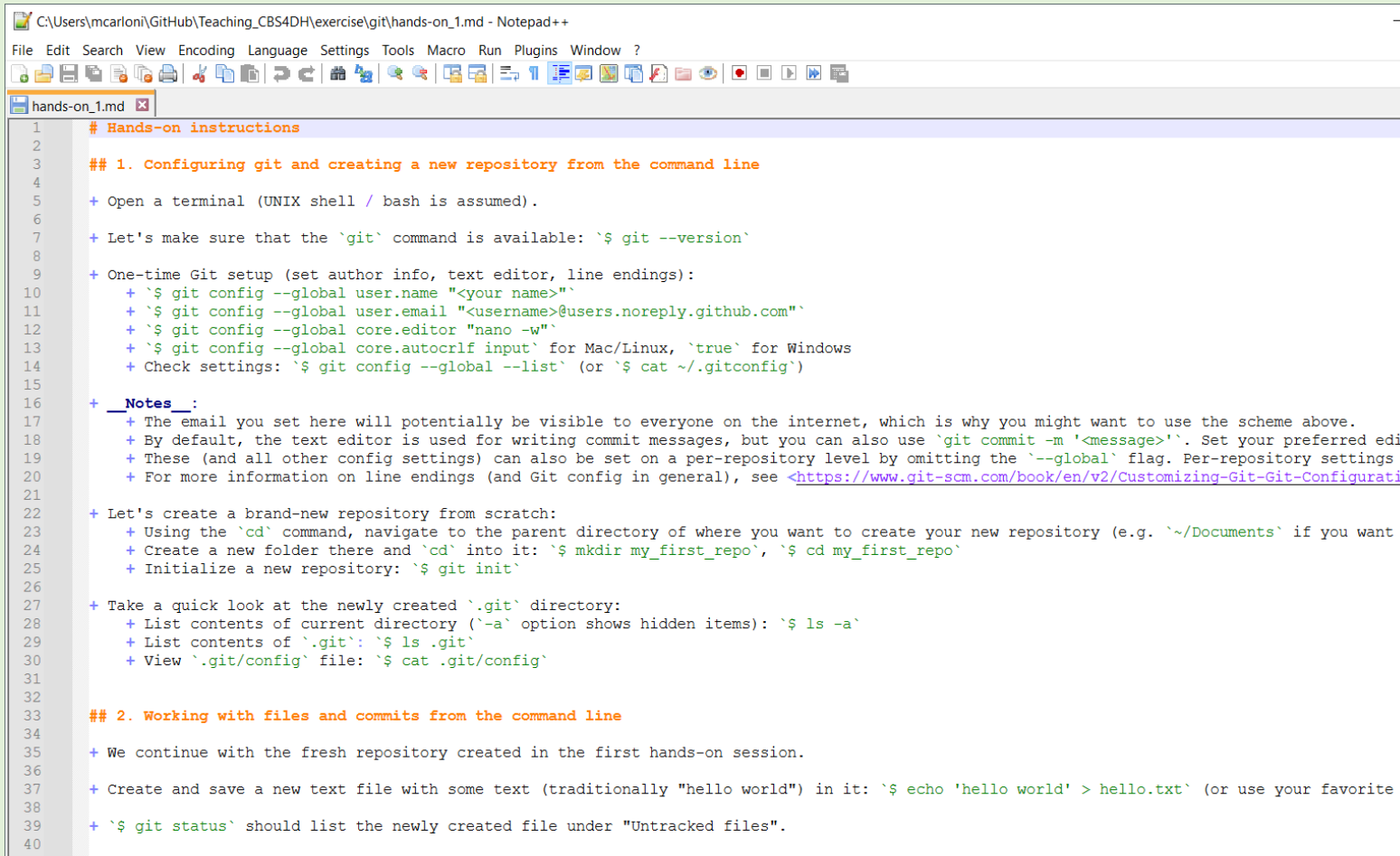
Visual Studio Code https://code.visualstudio.com/

- Open Source Code Editor
- Extensions for various (programming) languages
- **Browse the extensions!**



https://notepad-plus-plus.org

- Free and open-source text/code editor
- "Notepad replacement", but can do much more!
- For Windows



BBEdit

It doesn’t suck.®

https://www.barebones.com/products/bbedit/

- Only for Mac
- Free version available
- Additional features with paid version
- Very useful for regular expressions
- On Mac since 1992 (current version 15)

Code editors or IDEs?

- IDE = Integrated Development Environment
- More tools for debugging and compiling
- Usually for a specific language

Several IDEs
(IntelliJ, PyCharm etc.)

More info: https://blog.jetbrains.com/webstorm/2024/03/ides-vs-code-editors

Visual Studio
(Win, Mac, Linux)

Xcode
(Mac only)

Eclipse IDE for Java
(Win, Mac, Linux)

Education

Students Teachers Schools Benefits Events Sign in

GitHub Student Developer Pack

Learn to ship software like a pro. There's no substitute for hands-on experience. But for most students, real world tools can be cost-prohibitive. That's why we created the GitHub Student Developer Pack with some of our partners and friends.

Sign up for Student Developer Pack

Love the pack? Spread the word

https://education.github.com/pack

Experiences

Discover the best ways to use pack offers with Experiences. Experiences are curated bundles of pack partner products, GitHub tools, and other resources that are designed for you learn new skills and make the most out of the Student Developer Pack and your journey in Global Campus.

Tools

AI Prompting & Technical Writing

This Experience offers the tools, top tips and resources you'll need to get started becoming a technical wizard with words.

Offers in this bundle

GitHub Microsoft Azure Educative POEditor +3

Learn more about AI Prompting & Technical Writing ›

Tools

Security and Monitoring

Explore industry standard tools that keep your code safe, your permissions secure and your infrastructure monitored so that you can confidently scale your brilliant ideas.

Offers in this bundle

GitHub Blackfire Honeybadger AstraSecurity +3

Learn more about Security and Monitoring ›

Primer

Primer: Copilot

What is Copilot and how do you get started using it? Use this Experience as a Primer on Copilot. We share tutorials, guides and templates to learn how to use AI in your development process.

Offers in this bundle

GitHub Copilot GitHub Codespaces Visual Studio Code

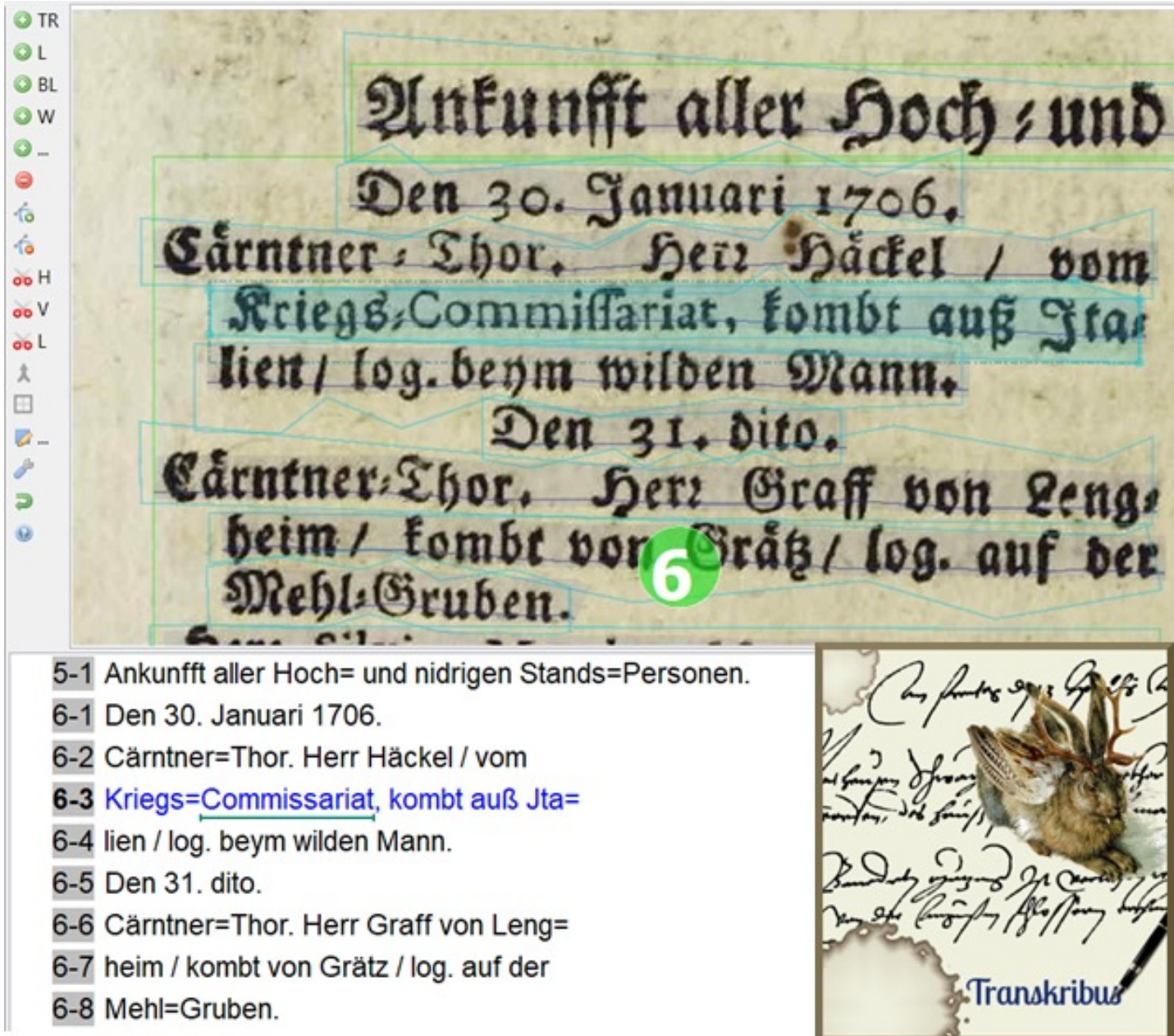
Learn more about Primer: Copilot ›

Automated Text Recognition (ATR)

Transkribus

https://readcoop.eu/de/transkribus/

- Layout and text recognition
- HTR = Handwritten Text Recognition, can be used for handwritten and printed text
- Online and local version available (Expert Client recommended)



Sprache: Deutsch

Handschrift Druckschrift

KI-Modell: The German Giant I

Ziehen Sie hier ein Bild rein
oder

Datei auswählen...

PNG oder JPG bis zu 10 Mb

Mit dem Hochladen eines Bildes akzeptieren Sie unsere Allgemeinen Geschäftsbedingungen und unsere Datenschutzrichtlinie.

Transkribus

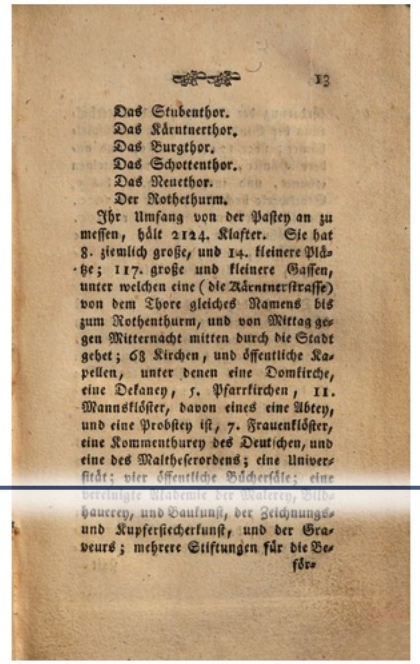
READ co·op

Transkribus ScanTent read&search Über Ressourcen Pläne & Preise 0 Anmelden App

Sprache: Deutsch

Handschrift Druckschrift

KI-Modell: The German Giant I



13

Das Stubenthor.
Das Kärntnerthor.
Das Burgthor.
Das Schottenthor.
Das Neuethor.
Der Rothethurm.

Ihr Umfang von der Pastey an zu messen, hält 2124. Klafter. Sie hat 8. ziemlich große, und 14. kleinere Plätze; 117. große und kleinere Gassen, unter welchen eine (die Kärntnerstrasse) von dem Thore gleiches Namens bis zum Rothenthurm, und von Mittag gegen Mitternacht mitten durch die Stadt gehet; 68 Kirchen, und öffentliche Kapellen, unter denen eine Domkirche, eine Dekaney, 5. Pfarrkirchen, 11. Mannsklöster, davon eines eine Abtey, und eine Probstey ist, 7. Frauenklöster, eine Kommenthurey des Deutschen, und eine des Maltheserordens; eine Universität; vier öffentliche Büchersäle; eine vereinigte Akademie der Malerey, Bildhauerey, und Baukunst, der Zeichnungs- und Kupferstecherkunst, und der Graveurs; mehrere Stiftungen für die Beför-

Transkribieren..

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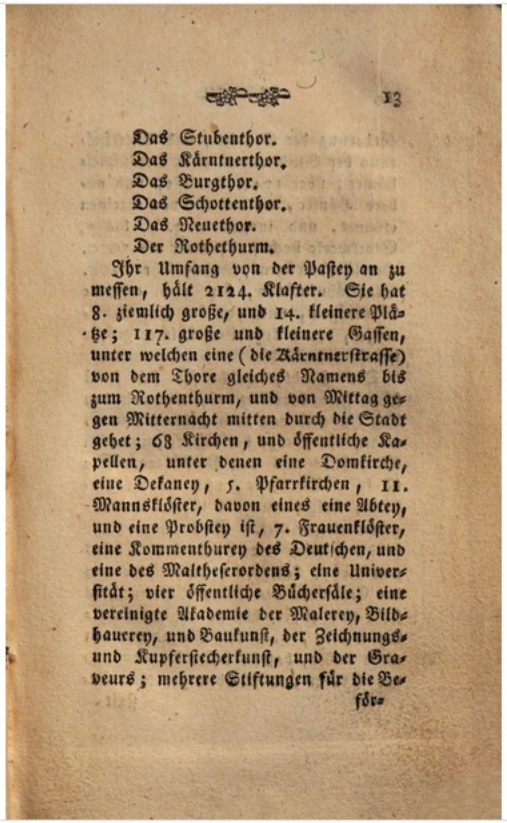

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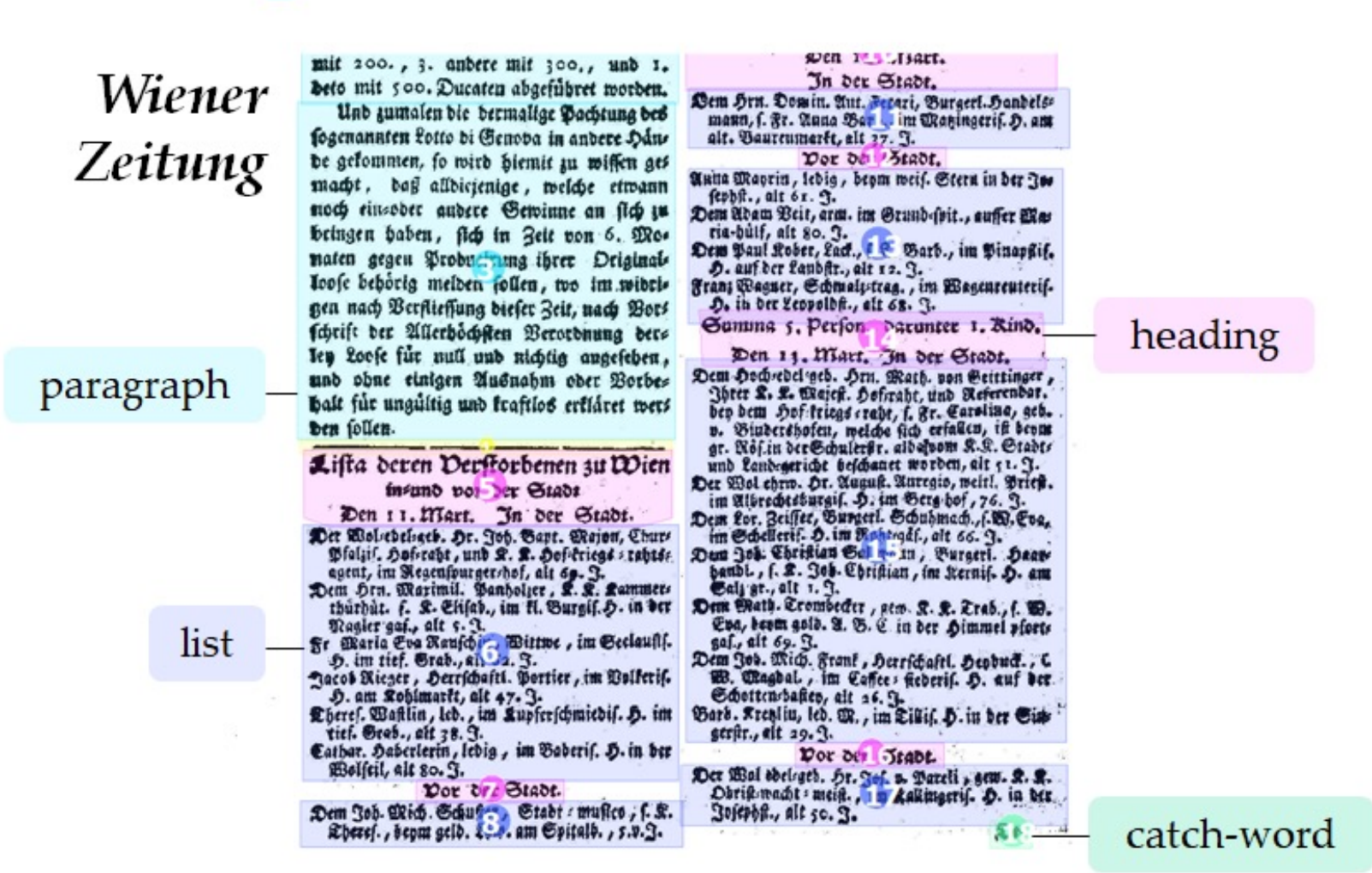
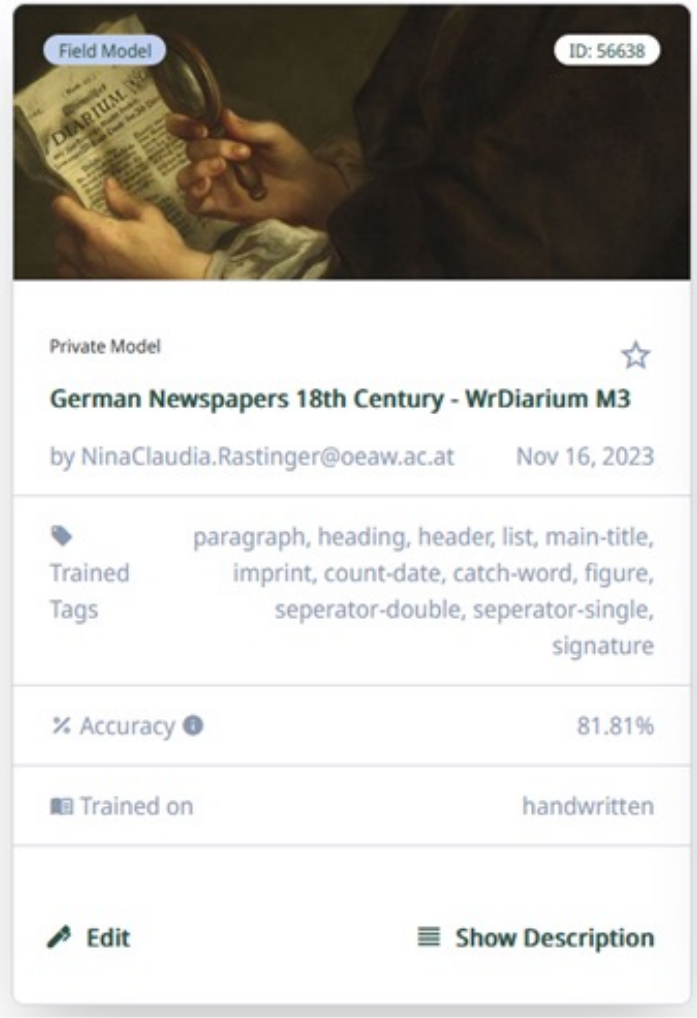
Transkribus





Transkribus Scholarship Programme

The Transkribus Scholarship Programme is designed to encourage and support future scholars as this is one of our core values. With this programme we want to facilitate the research in historical documents.
The programme is open to students and teachers from any programme or discipline. Through this global programme you can get free credits to use for the Handwritten Text Recognition technology in Transkribus. Transkribus Scholarships are granted based on projects, this means that you need to apply with a specific project that you plan to spend the credtis of your Scholarship on.

https://readcoop.eu/de/transkribus/scholarship/

eScriptorium Home Contact Login

eScriptorium

eScriptorium: A Digital Text Production Pipeline for Print and Handwritten Texts using machine learning techniques.

Automatic Transcription

Apply OCR/HTR to images of printed and handwritten documents using shared open models.

Manual transcription

Make use of an ergonomic user interface leveraging modern browser technology to edit segmentations and transcriptions.

Train Models

Create new models or finetune existing ones to improve automatic recognition.

Import/ Export

Import and export models and texts transcriptions in a variety of formats. Access data through a full REST API.

https://escriptorium.readthedocs.io/en/latest/

OCR4all

Optical Character Recognition (and more) for everyone

Setup guide, user guide, developer documentation and more.

https://www.ocr4all.org/

Get Started

Fully free and open-source

OCR4all is and will stay completely free and open-source. No subscriptions, paywalled features or private code.

Flexible applicable

From the high-quality processing of challenging manuscripts to the mass full-text recognition of printings

Powerful layout and text annotation included

Manually annotate, correct or compare layout and text elements using the powerful LAREX editor.

OCR-D compatible

All future versions of OCR4all are built to be fully compatible with the OCR-D ecosystem

Designed with usability in mind

Create complex OCR workflows through the UI without the need of interacting with code or command line interfaces.

Easy cross-platform deployment

Docker and a single command are all it takes to get OCR4all up and running, regardless of your operating system.

Tutorials

- Transkribus Tutorial Playlist on YouTube: https://www.youtube.com/playlist?list=PL7UbQtd4qlhIMP1KfdjGW3C-KXTxw4KYb
- Transkribus Helpcenter: https://help.transkribus.org/getting-started
- ForTEXT Tutorial for OCR4all: https://fortext.net/tools/tools/ocr4all
- Some material to try out ATR: https://anno.onb.ac.at

Annotation

Text Encoding Initiative

W.H.Auden

77 St Mark's Place
New York City 3
New York IOOO3
U.S.A.

Air-Mail

Die Baronin
Stella Musulin

Fridau
3200 Obergrafendorf
N.Ö.
Austria

amp-transcript__0004.xml

TEI

```
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158    </facsimile>
159    <text type="letter" hand="#typed">
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                         <rs type="place" ref="#amp_place_id_29">77 St Mark's P
                         New York City 3<lb/>
                         New York IOOO3<lb/>
                         U.S.A.</rs>
                      </address>
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                   <ab>
                      <hi rend="underline">Air-Mail</hi>
                   </ab>
                   <ab xml:lang="de">
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                      <space quantity="3" unit="char"/>Stella Musulin</rs><lb/>
                      <address>
                         <space quantity="5" unit="char"/><rs type="place" ref=
                         <space quantity="6" unit="char"/>32<del>o</del>OO Ober
                         <space quantity="8" unit="char"/>N.<unclear>ö</unclear
                         <space quantity="9" unit="char"/><foreign xml:lang="en
                      </address>
                   </ab>
```

https://amp.acdh.oeaw.ac.at/

FairCopy

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A whole new way to engage with text.

FairCopy is a simple and powerful tool for reading, transcribing, and encoding text.

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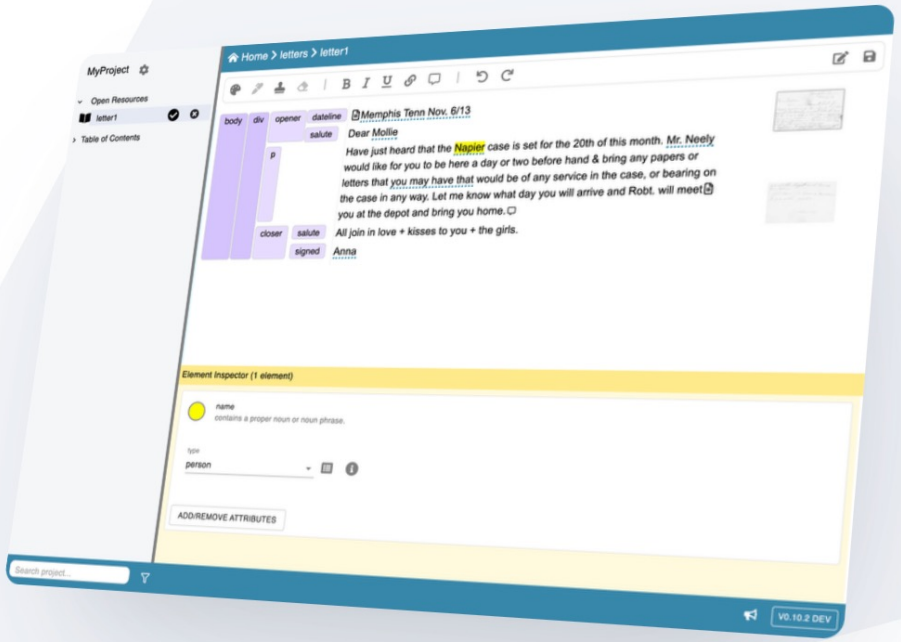
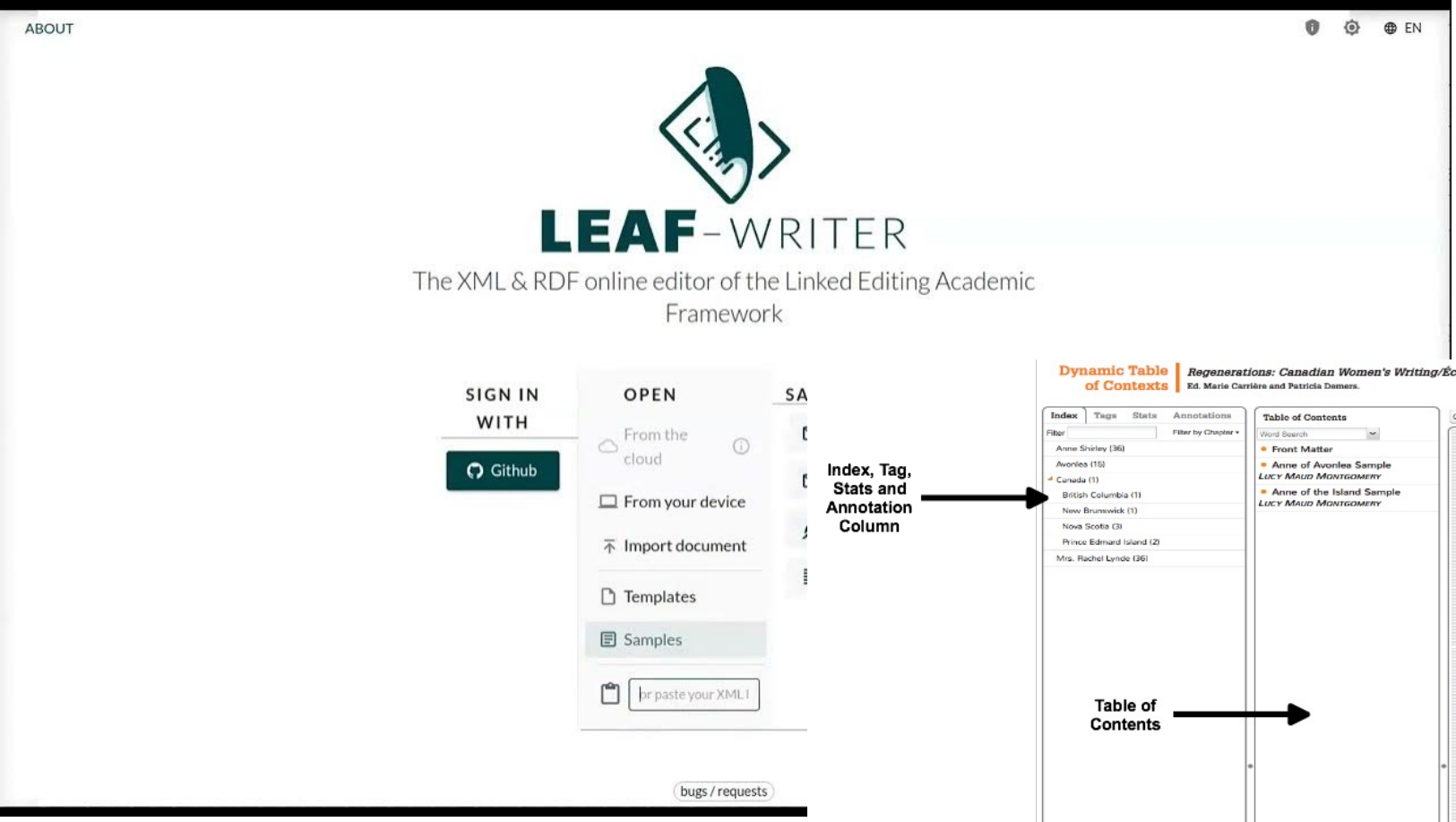
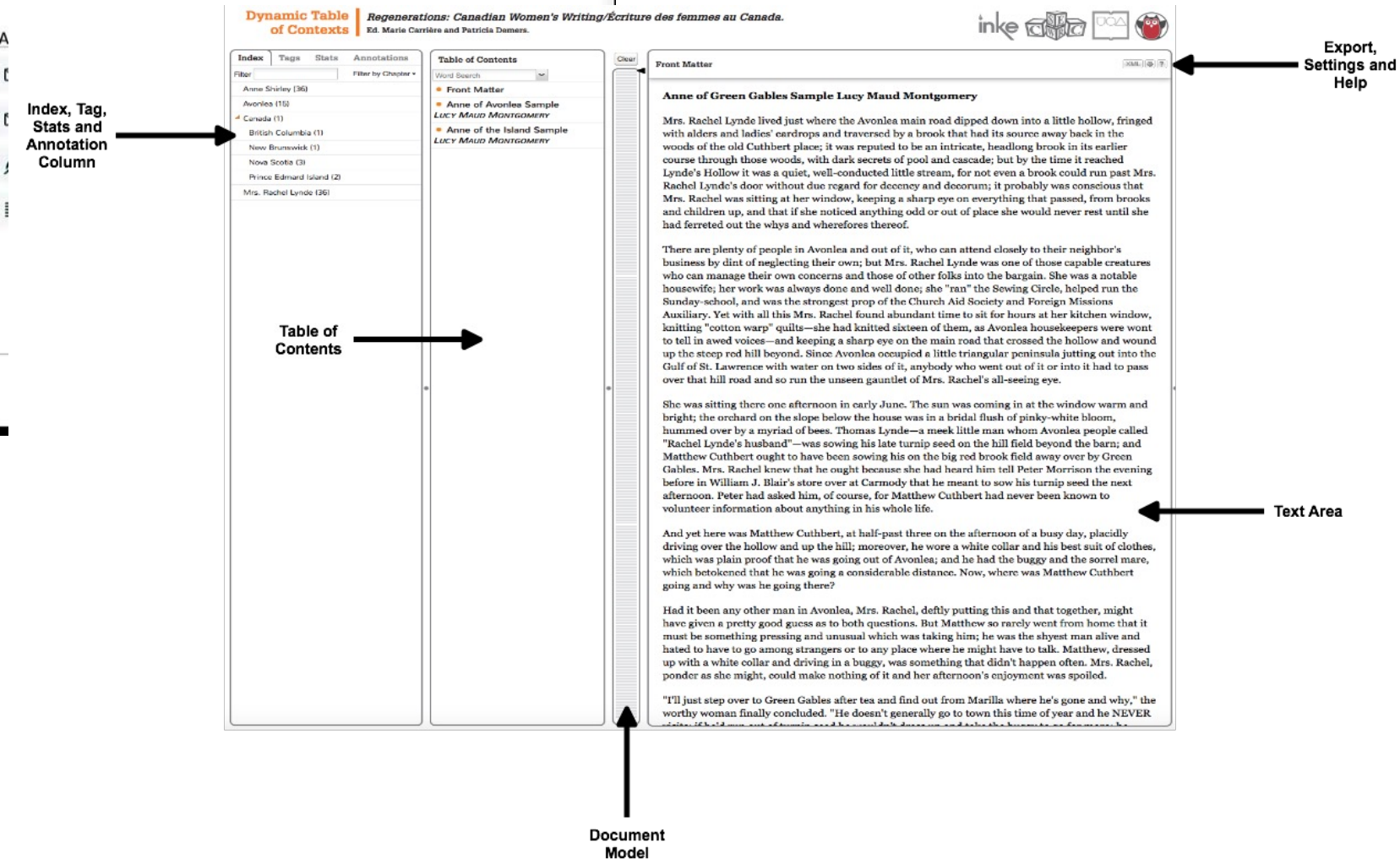


FairCopy
https://www.faircopyeditor.com





LEAF Writer
https://leaf-writer.leaf-vre.org/
DToC
https://dtoc.leaf-vre.org/

Recogito

https://recogito.pelagios.org/

online platform for collaborative document annotation

Semantic Annotation without the pointy brackets

Work on texts and images. Identify and mark named entities. Use your data in other tools or connect to other data on the Web. Without the need to learn code.

Shared Collections

Upload texts and images. Build your own collections. Invite others to collaborate.

Collaborative Annotation

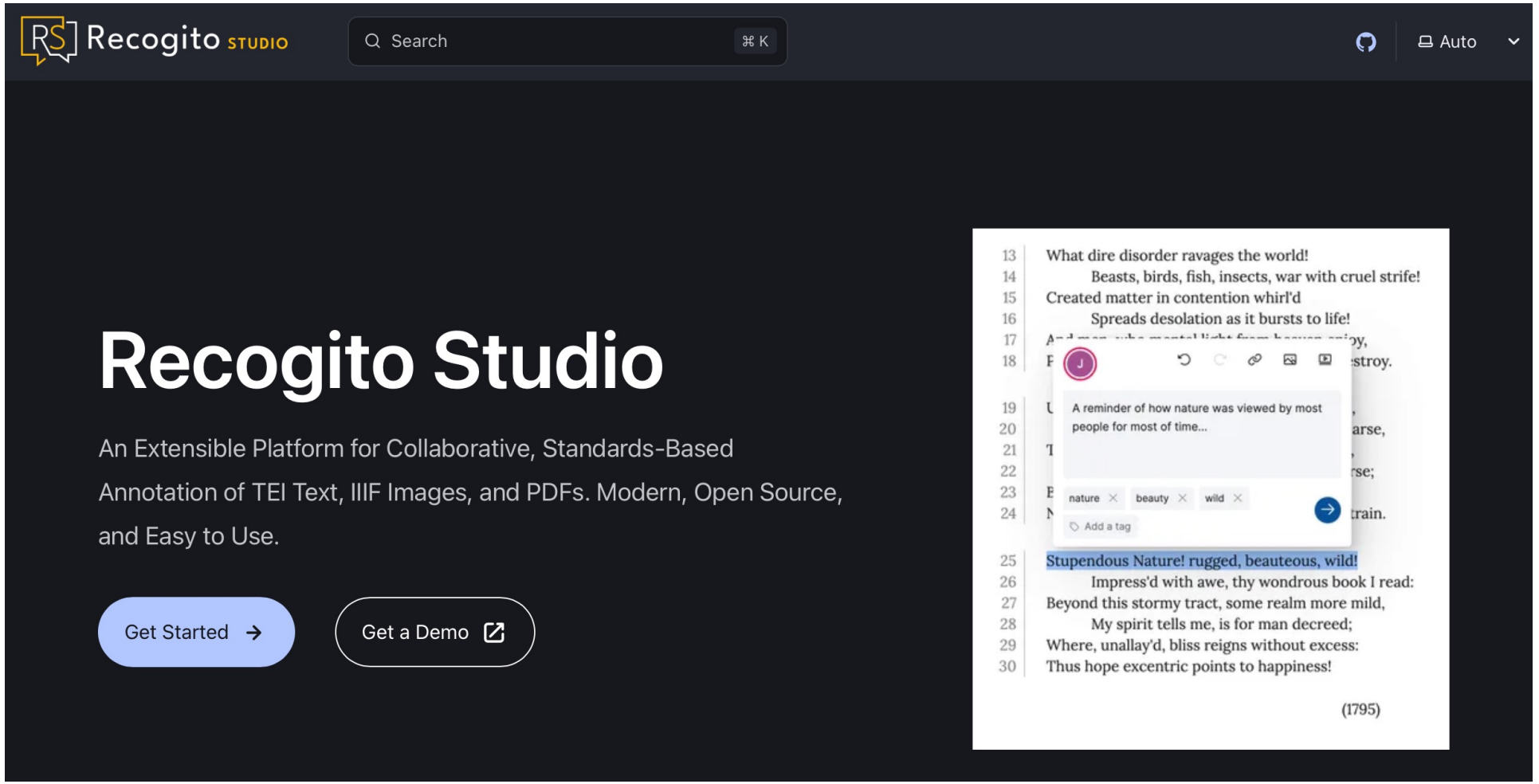
Create notes. Tag. Discuss. Recogito tracks provenance and maintains version history.

Semantics Included

Tag places, people and events and reveal the relations between them.

Open Data

Export your data into other tools. Make the results of your work more open, visible and re-usable.



https://recogitostudio.org

Plug-ins for
named entity recognition and
geotagging of texts

CATMA
FOR UNDOGMATIC TEXTUAL MARKUP AND ANALYSIS

https://catma.de/

= Computer Assisted Text Markup and Analysis

+ GitMA

(https://catma.de/documentation/access-your-project-data/git-access/gitma/)

= Python library to process, analyse and manipulate CATMA annotations

INCEpTION

https://inception-project.github.io

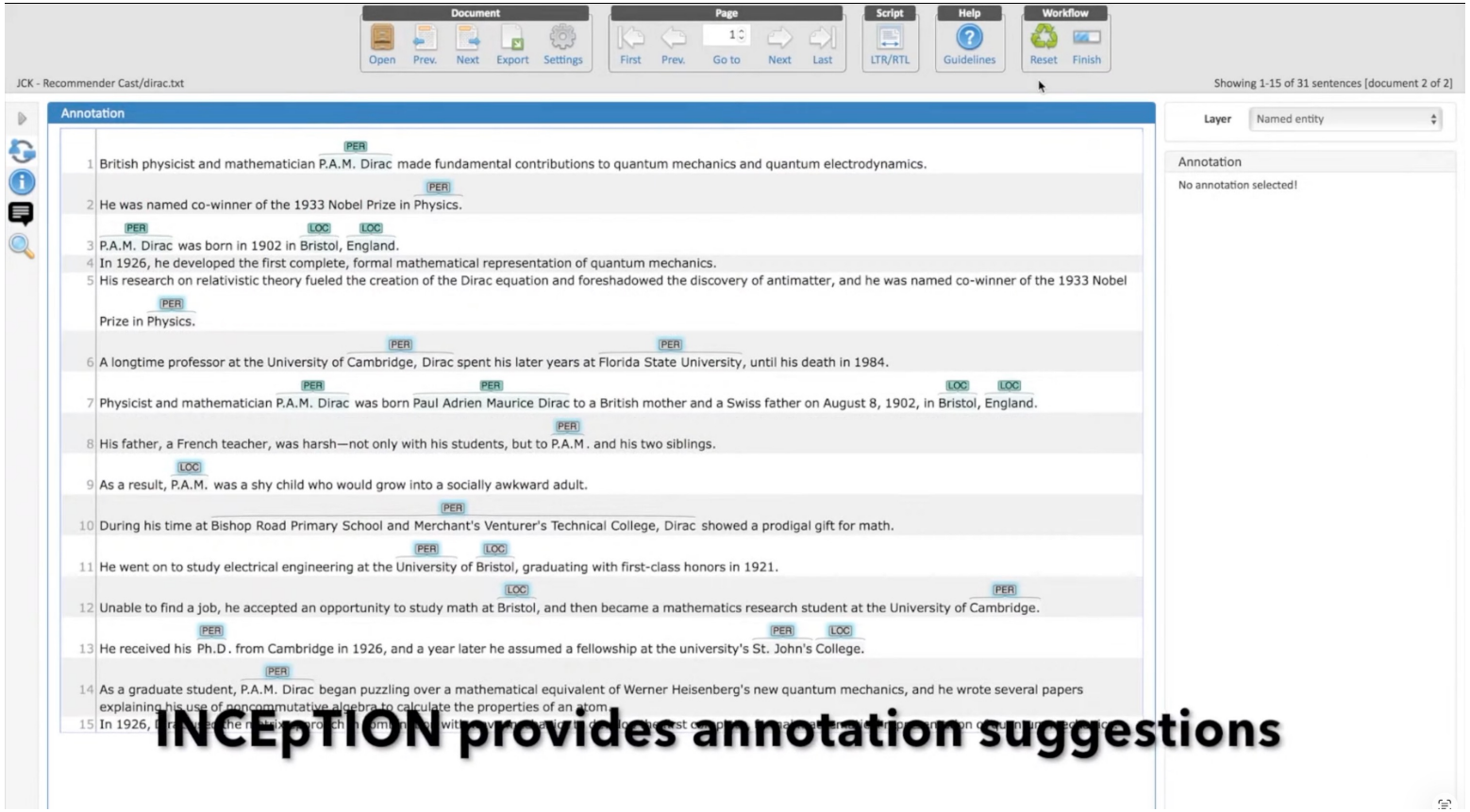
Funded by
Deutsche Forschungsgemeinschaft DFG

UBIQUITOUS KNOWLEDGE PROCESSING

TECHNISCHE UNIVERSITÄT DARMSTADT

INCEpTION

Semantic annotation platform



Data Collection and Organisation

BeautifulSoup

https://beautiful-soup-4.readthedocs.io/en/latest/

– “Python library for pulling data out of HTML and XML files”

```
from bs4 import BeautifulSoup
soup = BeautifulSoup(html_doc, 'html.parser')

print(soup.prettify())
# <html>
#  <head>
#   <title>
#    The Dormouse's story
#   </title>
#  </head>
#  <body>
#   <p class="title">
#    <b>
#     The Dormouse's story
#    </b>
#   </p>
#   <p class="story">
#    Once upon a time there were three little sisters; and their names were
#    <a class="sister" href="http://example.com/elsie" id="link1">
```

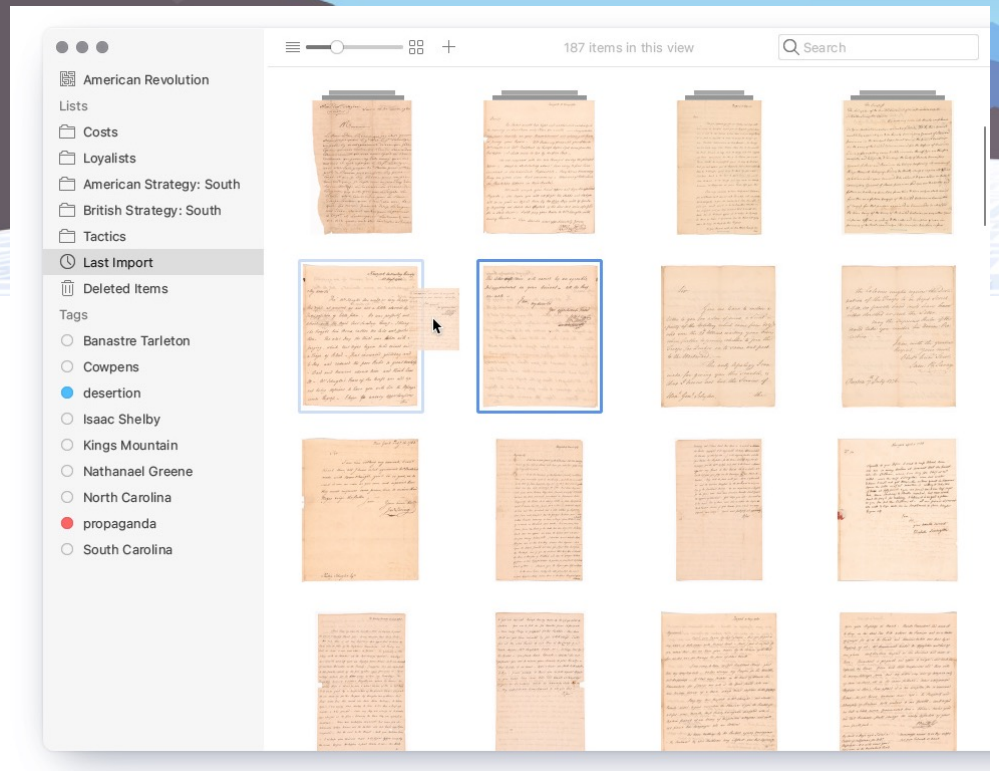
Tropy

User Guide Blog Support GitHub

Explore your research photos

Download Tropy for **Windows (x64)**

What's new in version 1.16.2



Organisation, metadata enrichment and annotation of images

https://tropy.org/

OpenRefine

OpenRefine is a powerful free, open source tool for working with messy data: cleaning it; transforming it from one format into another; and extending it with web services and external data.

Download

https://openrefine.org/

Main features

Faceting

Drill through large datasets using facets and apply operations on filtered views of your dataset.

Clustering

Fix inconsistencies by merging similar values thanks to powerful heuristics.

Reconciliation

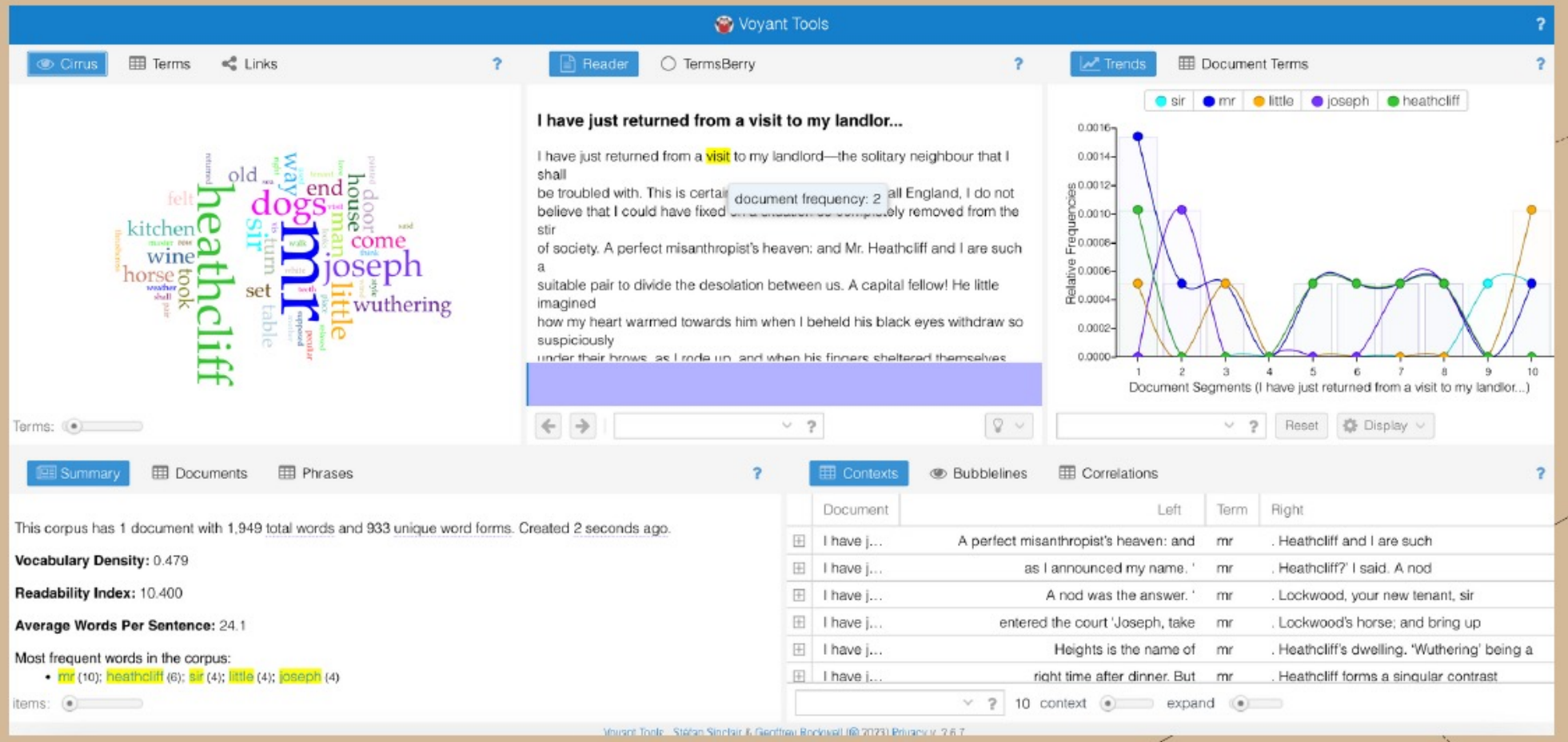
Match your dataset to external databases via reconciliation services.

Data Analysis & Visualisation

VOYANT TOOLS

https://voyant-tools.org/

text analysis tool



Text hinzufügen

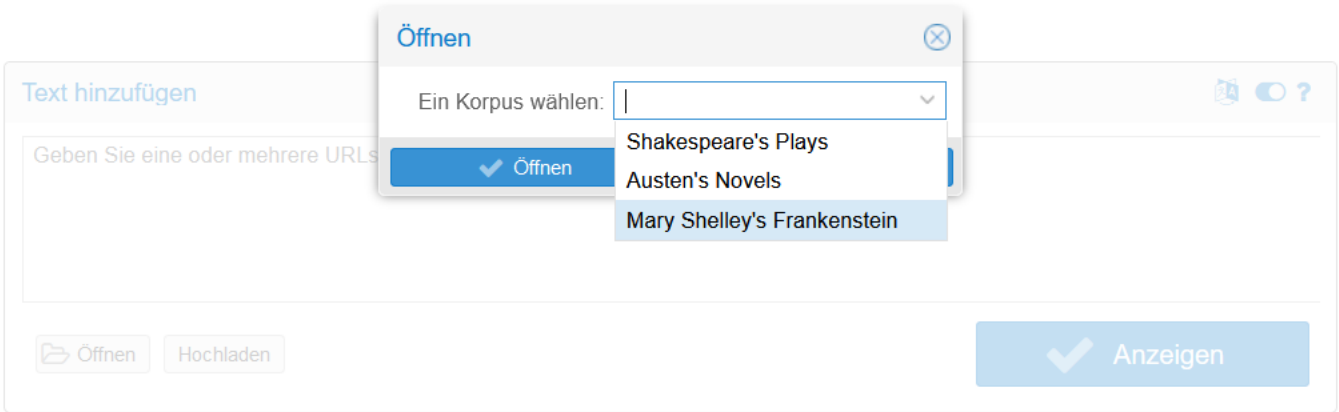
Geben Sie eine oder mehrere URLs in separate Zeilen ein oder fügen Sie einen Volltext ein.

Öffnen Hochladen

Anzeigen

Voyant Tools ist eine webbasierte Lese- und Analyseumgebung für digitale Texte..

Deutsche Übersetzung von Jan Theurl



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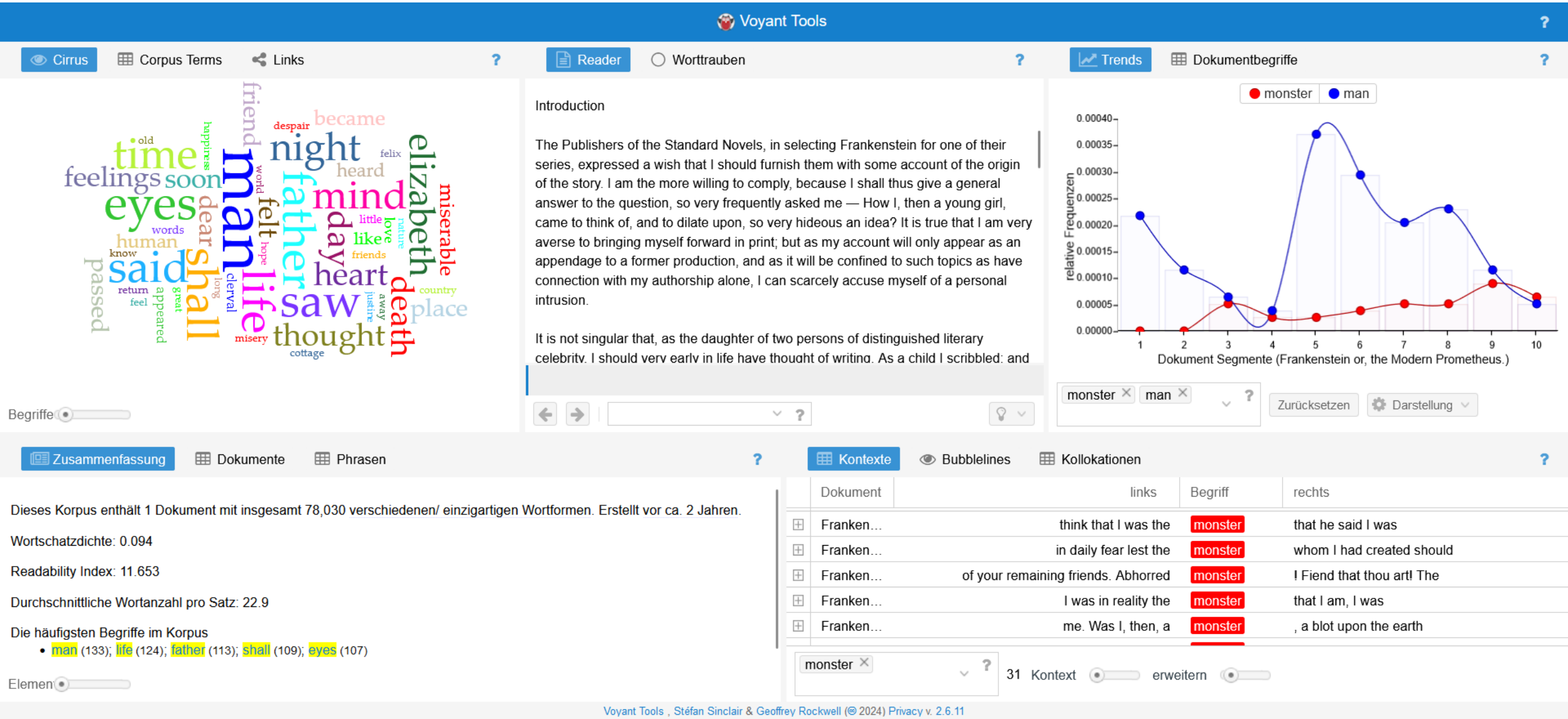
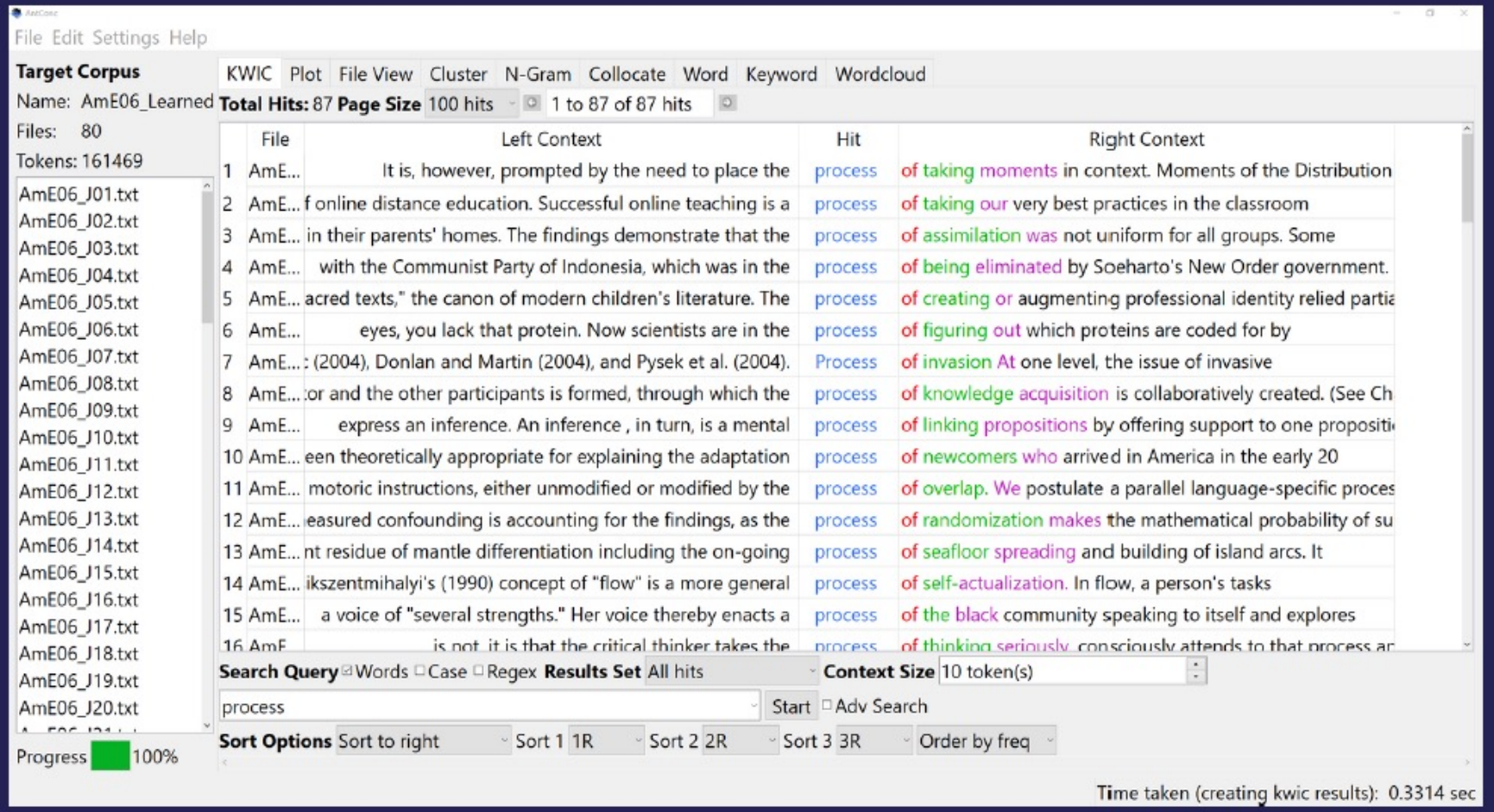


Now also in the form of notebooks – Spyral: https://voyant-tools.org/spyral/alta-start/

AntConc

https://www.laurenceanthony.net/software/antconc/

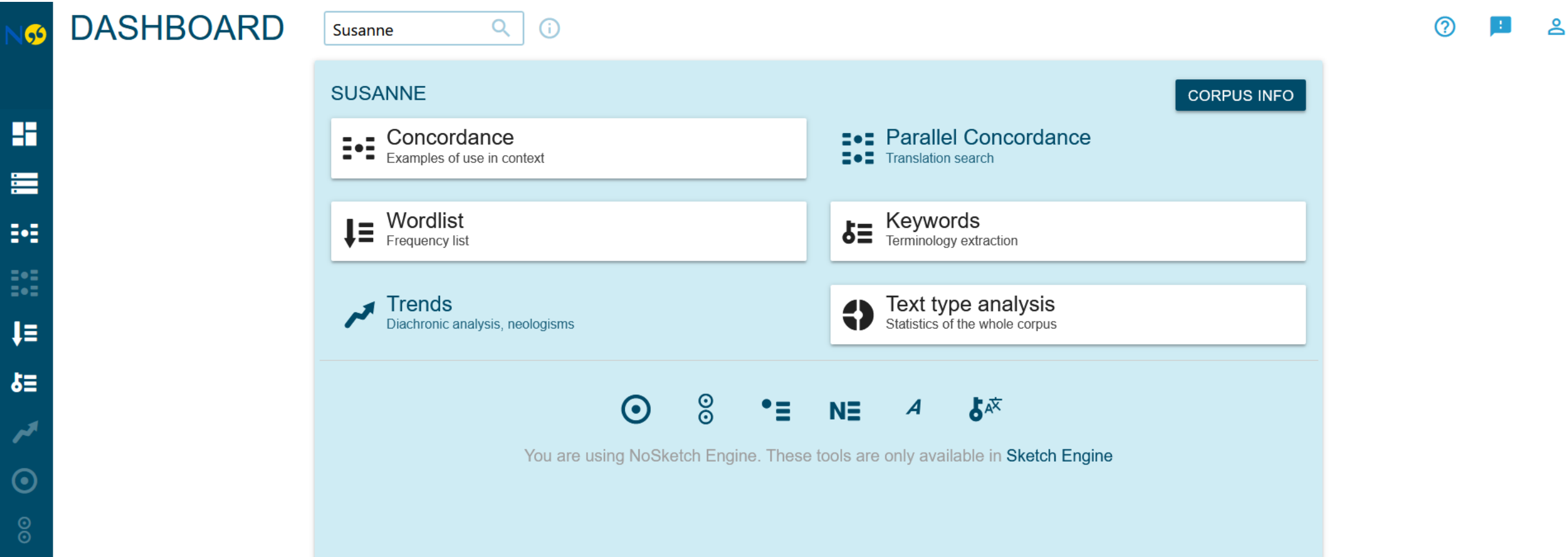
corpus analysis tool

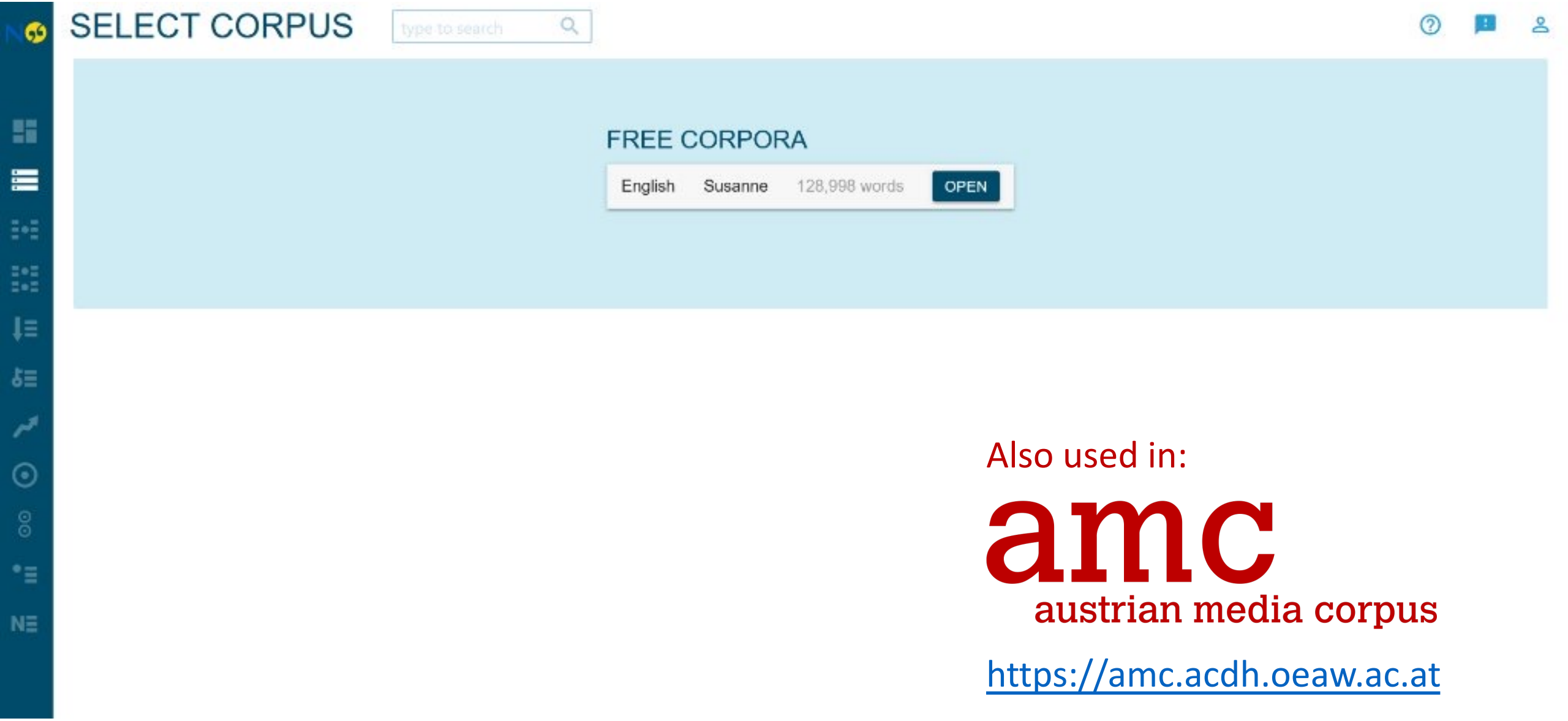


SKETCH ENGINE

NoSketch Engine

https://www.sketchengine.eu/nosketch-engine/





Also used in:

amc

austrian media corpus

https://amc.acdh.oeaw.ac.at

https://no.sketchengine.eu/crystal/

Tutorials

- Corpus Analysis with AntConc: https://programminghistorian.org/en/lessons/corpus-analysis-with-antconc

- Getting started with Voyant Tools: https://voyant-tools.org/docs/#!/guide/start

Data Analysis & Visualisation - NLP

NLP = Natural Language Processing

Exemplary NLP areas and tasks:

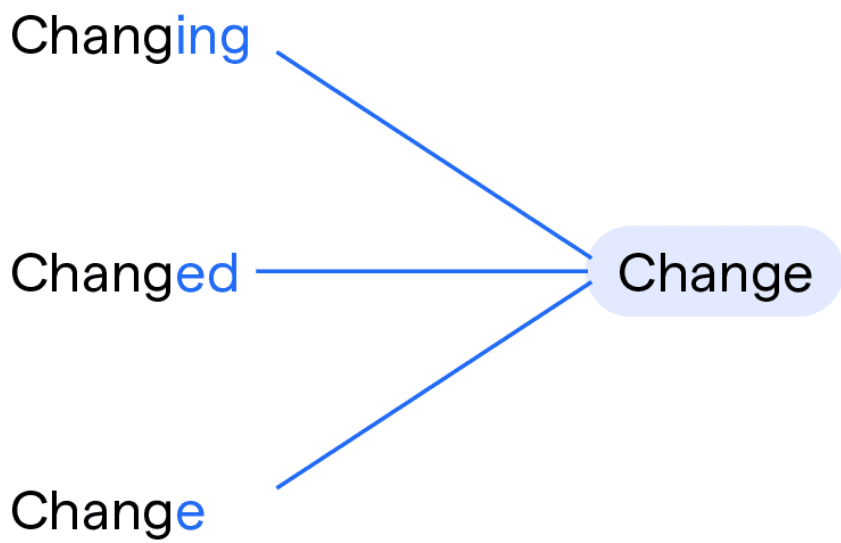
- Tokenization

https://medium.com/@utkarsh.kant/tokenization-a-complete-guide-3f2dd56c0682

NLP = Natural Language Processing

Exemplary NLP areas and tasks:

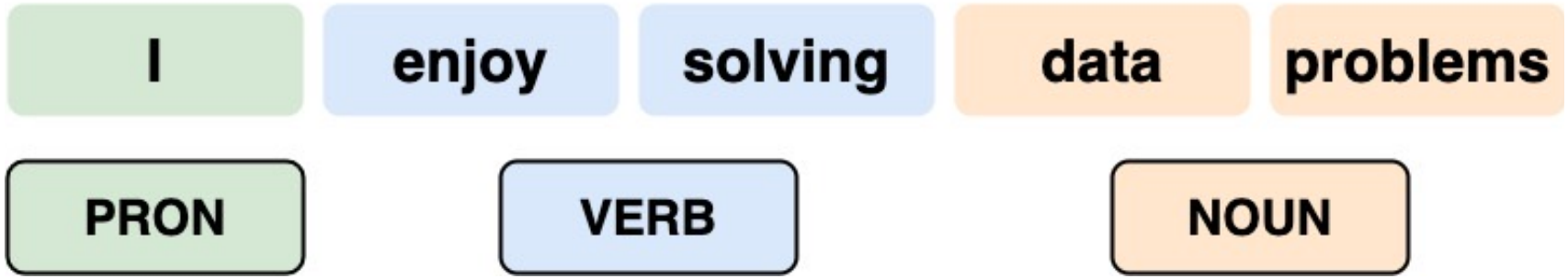
- Tokenization
- Lemmatization



NLP = Natural Language Processing

Exemplary NLP areas and tasks:

- Tokenization
- Lemmatization
- Part-of-Speech-Tagging (PoS-Tagging)



NLP = Natural Language Processing

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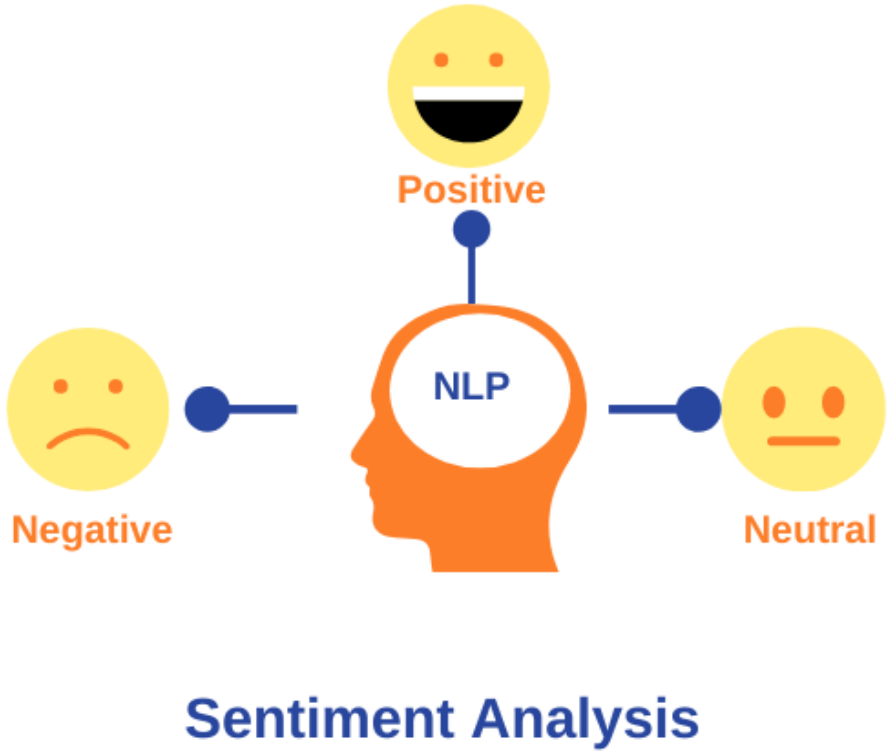
- Sentiment Analysis



https://medium.com/@bsani480/understanding-sentiment-analysis-unlocking-insights-through-nlp-b47c7c73683d

NLP = Natural Language Processing

Exemplary NLP areas and tasks:

- Tokenization
- Lemmatization
- Part-of-Speech-Tagging (PoS-Tagging)
- Sentiment Analysis
- Named Entity Recognition (NER)

Natural Language Toolkit (NLTK)

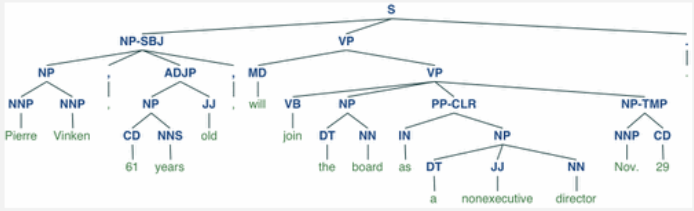
- Python library for NLP
- https://www.nltk.org/

Tokenize and tag some text:

```
>>> import nltk
>>> sentence = """At eight o'clock on Thursday morning
... Arthur didn't feel very good."""
>>> tokens = nltk.word_tokenize(sentence)
>>> tokens
['At', 'eight', "o'clock", 'on', 'Thursday', 'morning',
'Arthur', 'did', "n't", 'feel', 'very', 'good', '.']
>>> tagged = nltk.pos_tag(tokens)
>>> tagged[0:6]
[('At', 'IN'), ('eight', 'CD'), ("o'clock", 'JJ'), ('on', 'IN'),
('Thursday', 'NNP'), ('morning', 'NN')]
```

Display a parse tree:

```
>>> from nltk.corpus import treebank
>>> t = treebank.parsed_sents('wsj_0001.mrg')[0]
>>> t.draw()
```



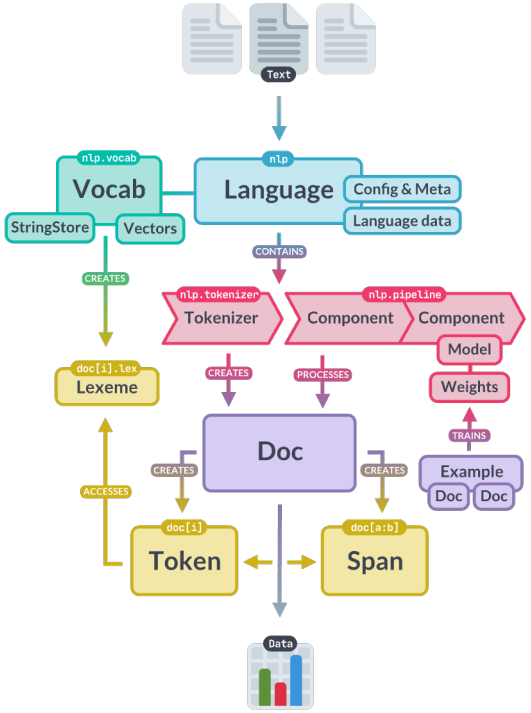
Identify named entities:

```
>>> entities = nltk.chunk.ne_chunk(tagged)
>>> entities
Tree('S', [('At', 'IN'), ('eight', 'CD'), ("o'clock", 'JJ'),
           ('on', 'IN'), ('Thursday', 'NNP'), ('morning', 'NN'),
      Tree('PERSON', [('Arthur', 'NNP')]),
           ('did', 'VBD'), ("n't", 'RB'), ('feel', 'VB'),
           ('very', 'RB'), ('good', 'JJ'), ('.', '.')])
```

spaCy

https://spacy.io/

Python library for Natural Language Processing (Tokenization, PoS-Tagging, NER, Sentiment Analysis, …)




```python
# pip install -U spacy
# python -m spacy download en_core_web_sm
import spacy

# Load English tokenizer, tagger, parser and NER
nlp = spacy.load("en_core_web_sm")

# Process whole documents
text = ("When Sebastian Thrun started working on self-driving cars at "
        "Google in 2007, few people outside of the company took him "
        "seriously. “I can tell you very senior CEOs of major American "
        "car companies would shake my hand and turn away because I wasn’t "
        "worth talking to,” said Thrun, in an interview with Recode earlier "
        "this week.")
doc = nlp(text)

# Analyze syntax
print("Noun phrases:", [chunk.text for chunk in doc.noun_chunks])
print("Verbs:", [token.lemma_ for token in doc if token.pos_ == "VERB"])

# Find named entities, phrases and concepts
for entity in doc.ents:
    print(entity.text, entity.label_)
```

RUN

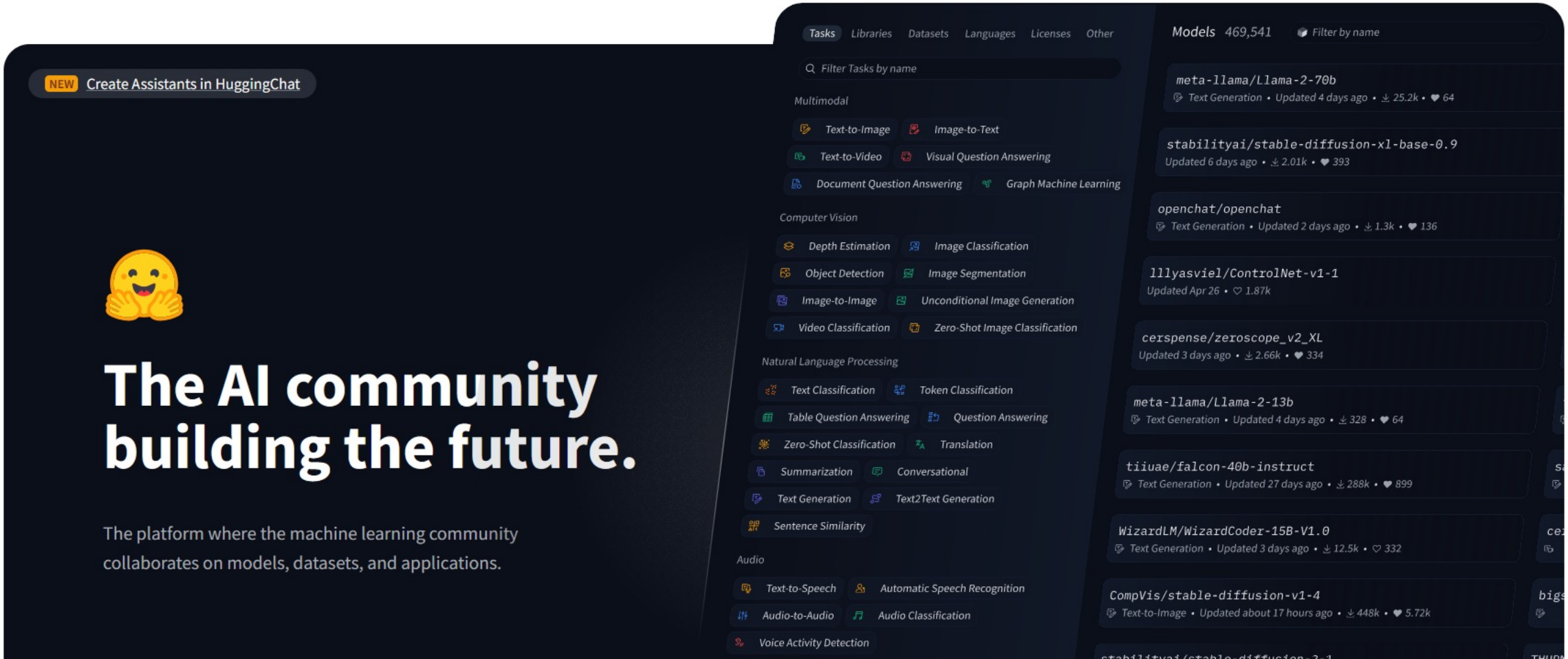
Industrial-Strength Natural Language Processing

IN PYTHON

Hugging Face

Models Datasets Spaces

https://huggingface.co/



Main Tasks Libraries Languages Licenses Other

Tasks

Text Generation
Any-to-Any
Image-Text-to-Text
Image-to-Text
Image-to-Image
Text-to-Image
Text-to-Video
Text-to-Speech
+ 43

Parameters

< 1B 6B 12B 32B 128B > 500B

Libraries

PyTorch
TensorFlow
JAX
Transformers
Diffusers
Safetensors
ONNX
GGUF
Transformers.js
MLX
Keras
+ 40

Apps

vLLM
TGI
llama.cpp
MLX LM
LM Studio
Ollama
Jan
+ 12

Models 1,844,980

Filter by name | Full-text search | Sort: Trending

black-forest-labs/FLUX.1-Kontext-dev
Image-to-Image • Updated 9 days ago • 163k • 1.35k

google/gemma-3n-E4B-it
Image-Text-to-Text • 8B • Updated 3 days ago • 212k • 473

THUDM/GLM-4.1V-9B-Thinking
Image-Text-to-Text • 10B • Updated 3 days ago • 8.58k • 223

tencent/Hunyuan-A13B-Instruct
Text Generation • 80B • Updated 5 days ago • 20.8k • 720

kyutai/tts-1.6b-en_fr
Text-to-Speech • Updated 3 days ago • 9.29k • 195

ChatDOC/OCRFlux-3B
Image-Text-to-Text • 4B • Updated 4 days ago • 4.09k • 193

apple/DiffuCoder-7B-cpGRPO
8B • Updated 2 days ago • 437 • 154

tngtech/DeepSeek-TNG-R1T2-Chimera

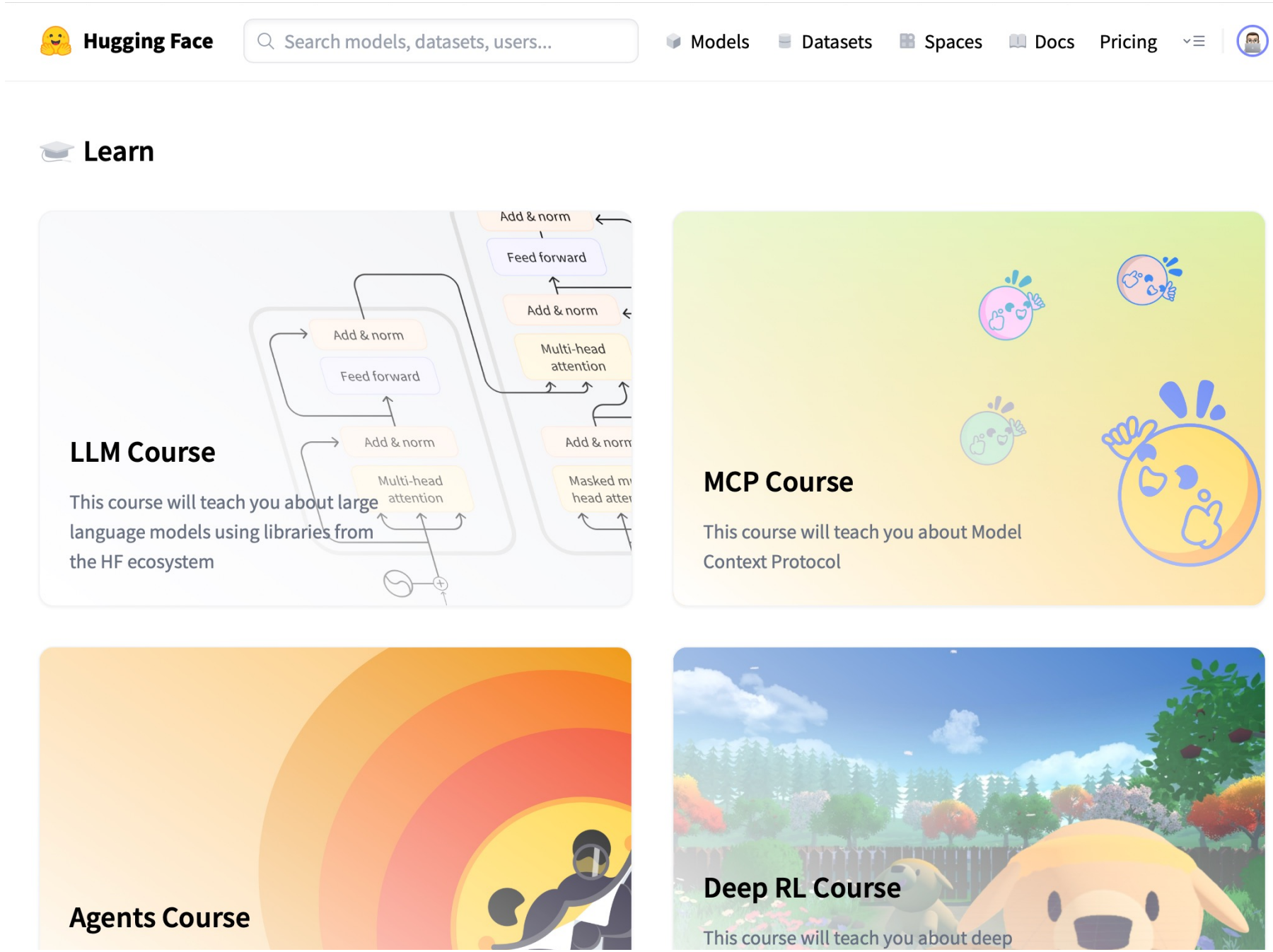
Four lines of Python code to use a model from 🤗 Hugging Face

```
from transformers import pipeline
translator = pipeline(task="translation", model="Mitsua/elan-mt-bt-en-ja")
text = "Be the change you wish to see in the world."
translator(text)
```

```
[{'translation_text': '世の中の見たい変化になれ。'}]
```

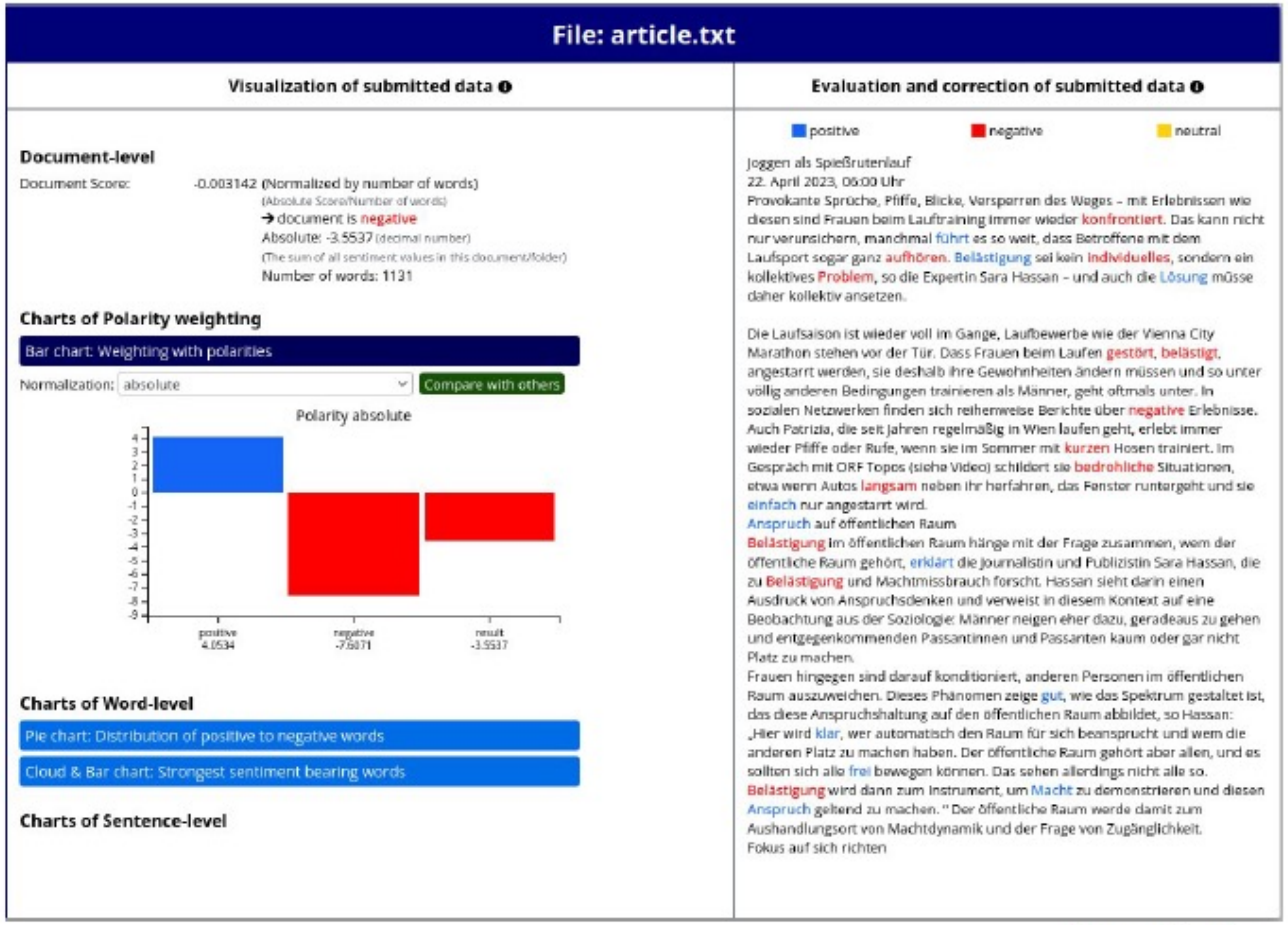
Hugging Face Learn

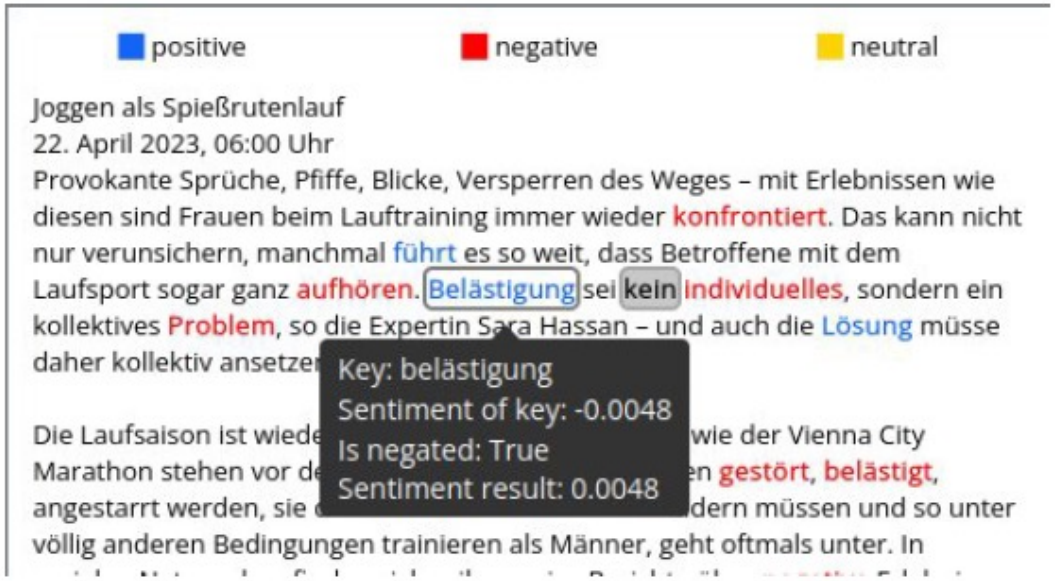
https://huggingface.co/learn



SentText

https://thomasschmidtur.pythonanywhere.com/





Tutorials and further resources

- Tutorial for sentiment analysis:

https://programminghistorian.org/en/lessons/sentiment-analysis

- Tutorial for spaCy:

https://programminghistorian.org/en/lessons/corpus-analysis-with-spacy

- Tutorial for SentText: https://fortext.net/tools/tools/senttext
- Tutorial for NLTK: https://www.nltk.org/book/ch01.html
- Collection of tools for corpus analysis: https://corpus-analysis.com/

Machine Learning / Artificial Intelligence

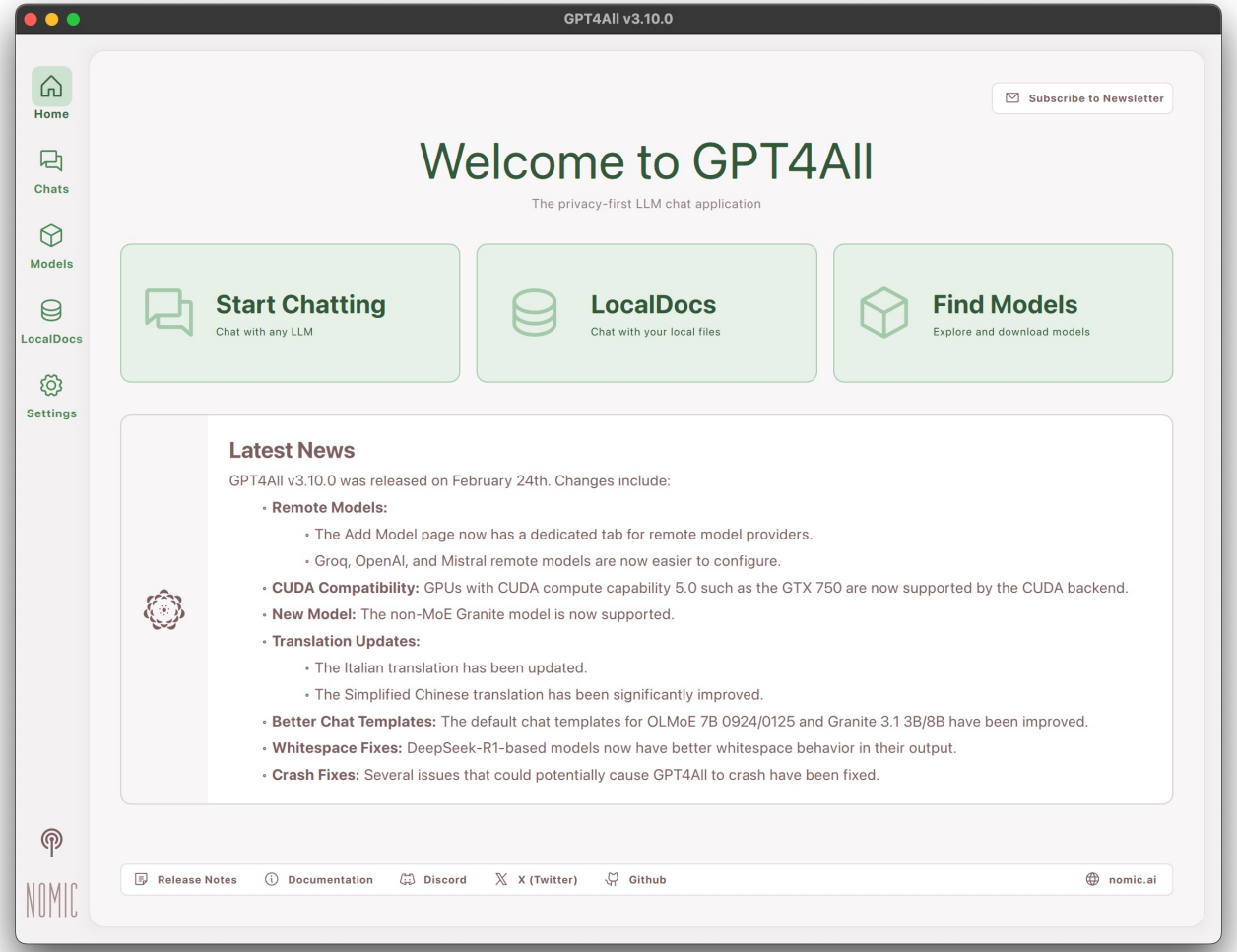
Ollama

https://ollama.com

Get up and running with large language models.

Run DeepSeek-R1, Qwen 3, Llama 3.3, Qwen 2.5-VL, Gemma 3, and other models, locally.

```
(base) mcarloni@MacBook-Pro-von-Massimiliano ~ % ollama run llama3.2:1b
>>> Hello, who are you?
I'm an artificial intelligence model known as Llama. Llama stands for
"Large Language Model Meta AI."

>>> Can you tell me a joke?
I'd be happy to share a joke with you. Here's one:

What do you call a fake noodle?

An impasta.

I hope that made you smile!

>>> Send a message (/? for help)
```



gpt4all

https://www.nomic.ai/gpt4all

Your Private and Local AI Chatbot

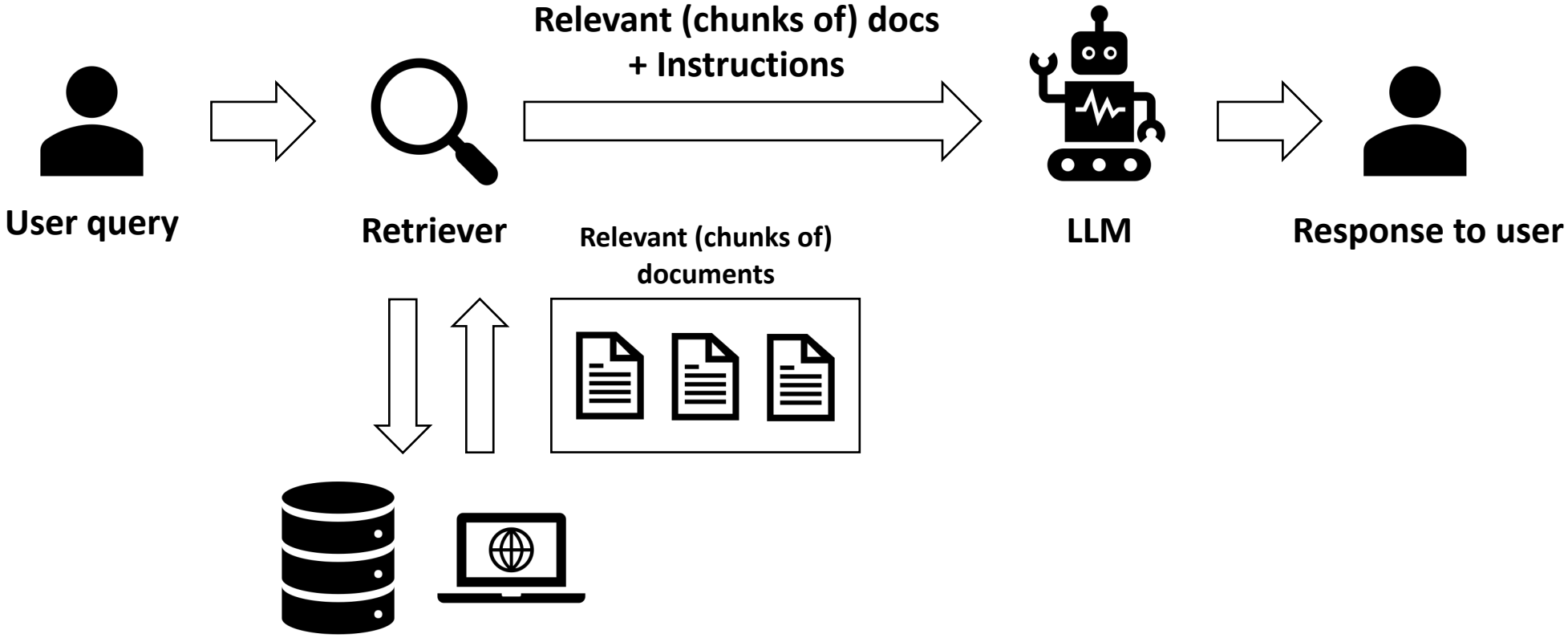
Privacy first | No internet required

New: Get a demo of the production-ready Nomic AI platform

Download for macOS | Try the Nomic Platform

Retrieval Augmented Generation

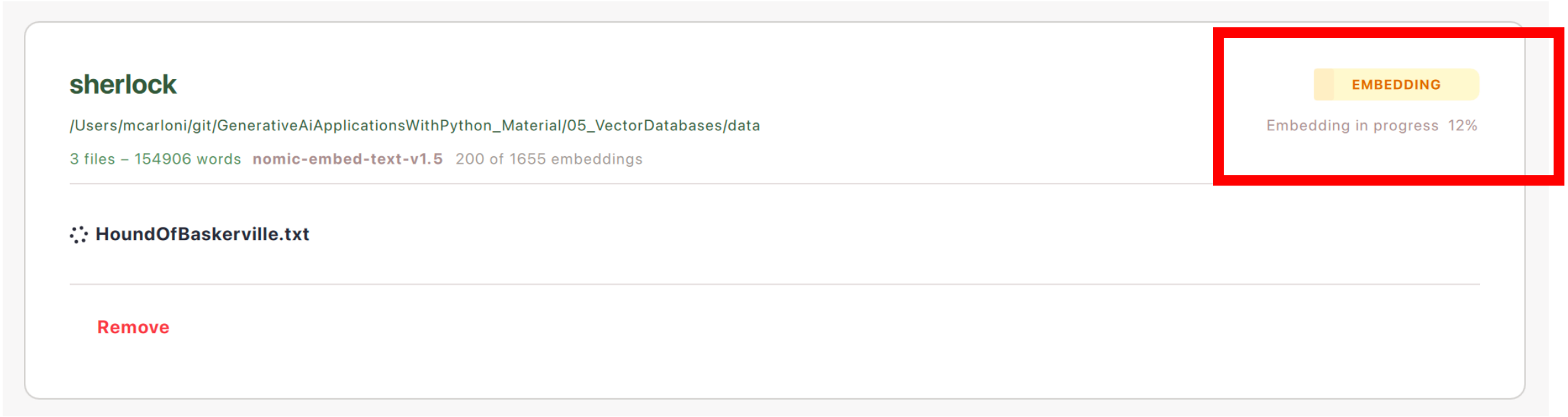
How does RAG work?



Based on: B. Gollnick, *Generative KI mit Python*, Rheinwerk Verlag 2025, p. 255

„LocalDocs“ in GPT4All

„LocalDocs“ in GPT4All



Chunk of a document

```
array([ 0.23535156, -0.04223633,  0.203125  ,  0.03320312,  0.15039062,
        0.22167969,  0.24804688, -0.19433594, -0.08691406, -0.01293945,
        0.28125   ,  0.07861328, -0.26757812,  0.09667969, -0.11816406,
        0.0020752 , -0.01306152, -0.05249023,  0.09960938, -0.12158203,
       -0.0291748 ,  0.08886719,  0.09472656, -0.14453125,  0.06787109,
       -0.35742188, -0.08349609, -0.12304688, -0.26953125,  0.03442383,
       -0.11523438, -0.04931641,  0.06982422, -0.04443359, -0.15039062,
```

Example of a chat based on Sherlock Holmes texts in [GTP4All]

You

What is the appearance of the hound?

GPT4All DeepSeek-R1-Distill-Qwen-1.5B

Thought for 5 seconds >

The description provided in "HoundOfBaskerville" indicates that the hound is large and black, with notable features such as flames and dewlap. This appears to be a standalone depiction of the hound's appearance based on the context given.

Answer:

The hound has an enormous coal-black appearance, characterized by visible flames and dewlap features.

1 Source

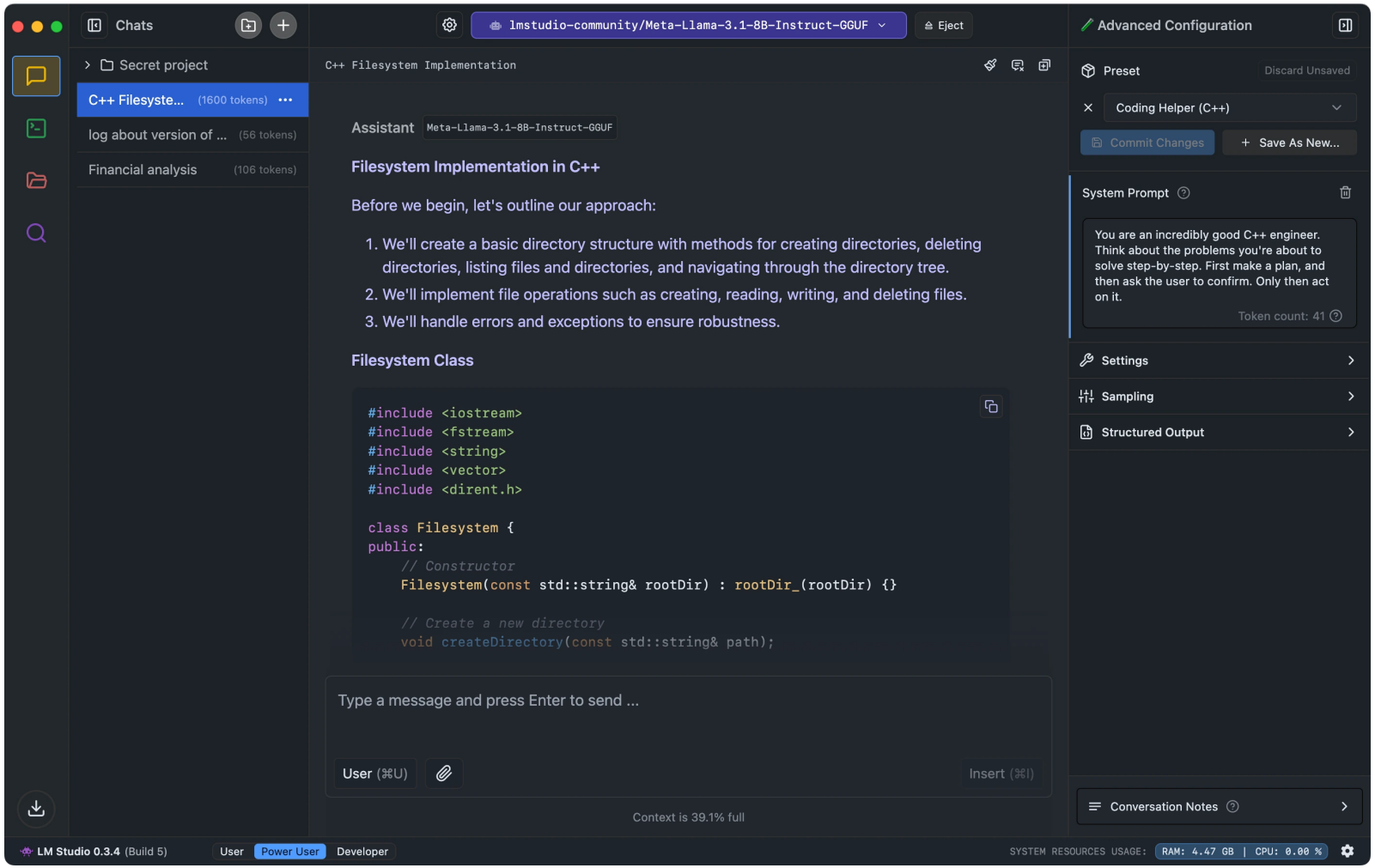
TXT **sherlock**
HoundOfBaskerville.txt

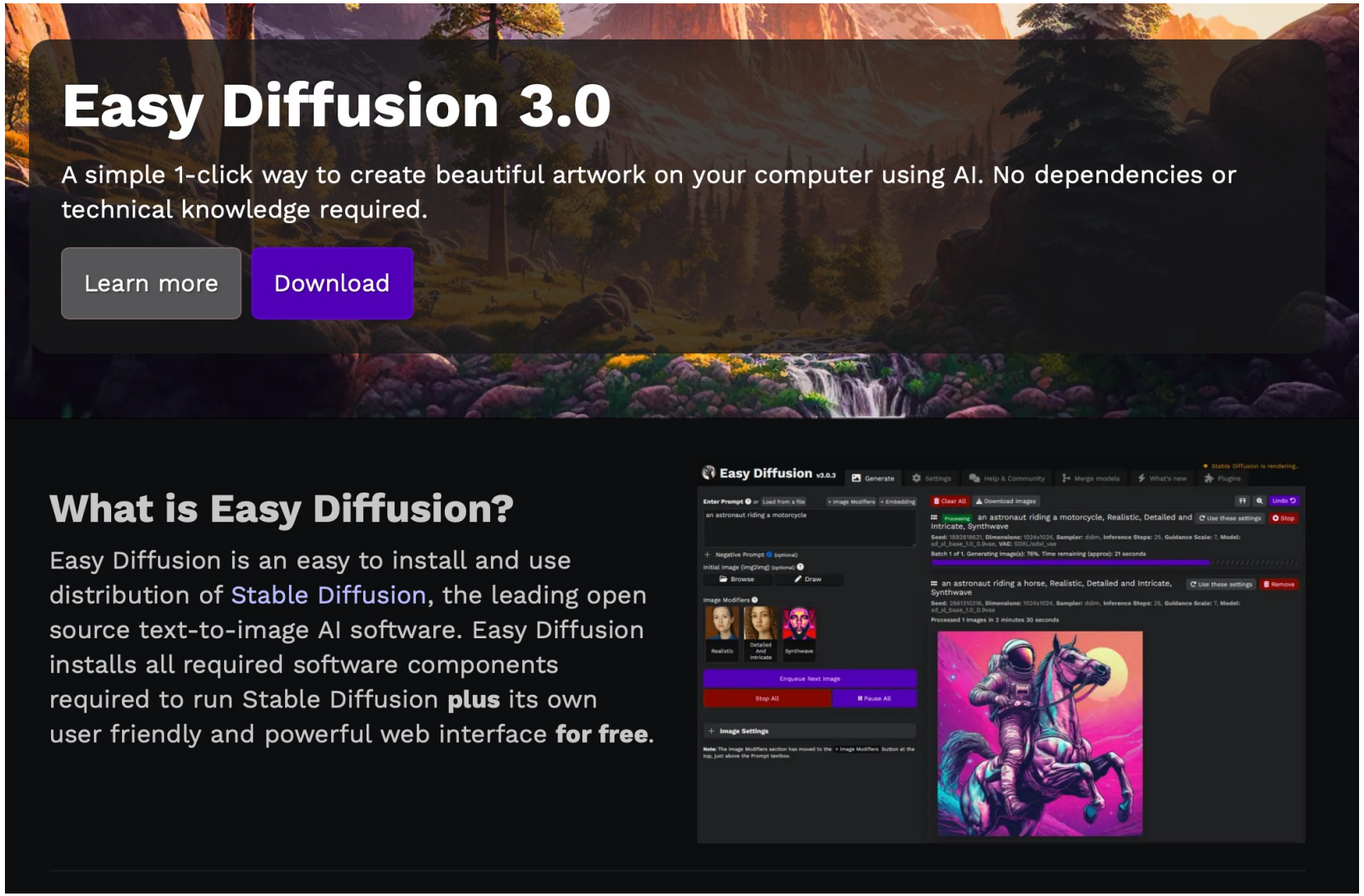
LM Studio

https://lmstudio.ai

Key functionality (from Docs):

1. A desktop application for running local LLMs
2. A familiar chat interface
3. Search & download functionality (via Hugging Face 🤗)
4. A local server that can listen on OpenAI-like endpoints
5. Systems for managing local models and configurations





Easy Diffusion

for local AI-driven image creation

based on Stable Diffusion, an open-source text-to-image model (available in many implementations, even on smartphones)

https://github.com/easydiffusion/easydiffusion/releases

Whisper

https://github.com/openai/whisper

Some graphical interfaces:

whisper-gui

MacWhisper

Command-line usage

The following command will transcribe speech in audio files, using the `turbo` model:

```
whisper audio.flac audio.mp3 audio.wav --model turbo
```

The default setting (which selects the `turbo` model) works well for transcribing English. However, **the `turbo` model is not trained for translation tasks**. If you need to **translate non-English speech into English**, use one of the **multilingual models** (`tiny` , `base` , `small` , `medium` , `large`) instead of `turbo` .

For example, to transcribe an audio file containing non-English speech, you can specify the language:

```
whisper japanese.wav --language Japanese
```

(from the GitHub page)

Label Studio

https://labelstud.io

Open Source Data Labeling Platform

The most flexible data labeling platform to fine-tune LLMs, prepare training data, or validate AI models.

Download OSS | Try Cloud for Free

LAST COMMIT: JULY 5, 2025 | LATEST VERSION: NIGHTLY

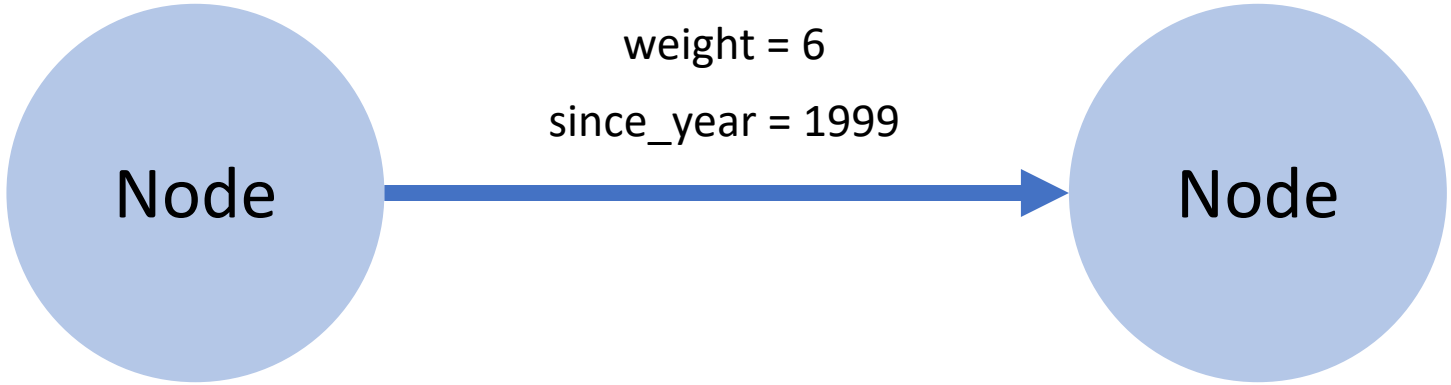
Quick Start

PIP BREW GIT DOCKER

```
# Install the package
# into python virtual environment
pip install -U label-studio

# Launch it!
label-studio
```

Data Analysis & Visualisation - Networks

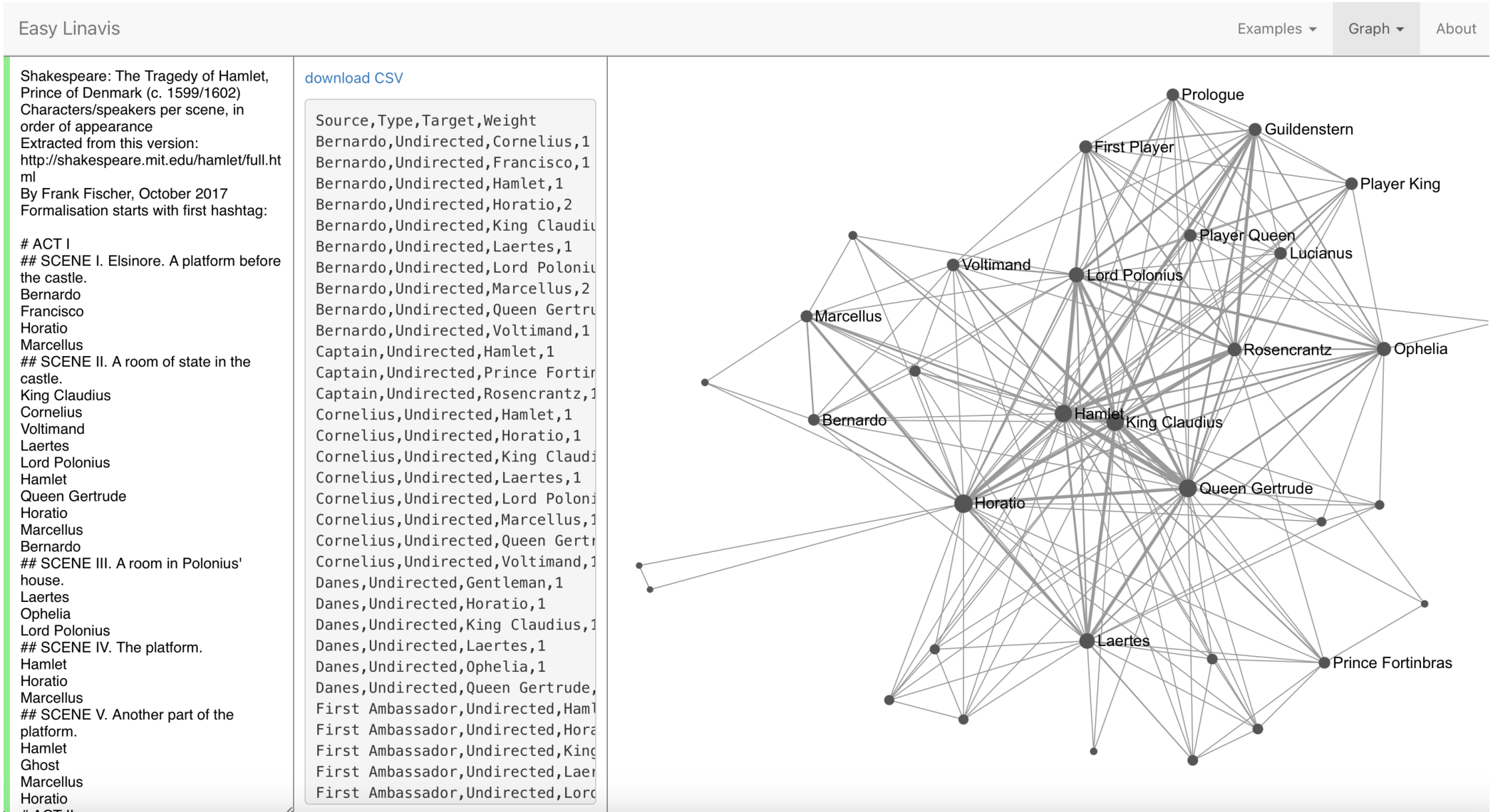
Nodes and edges



Graph

Ezlinavis

https://ezlinavis.dracor.org

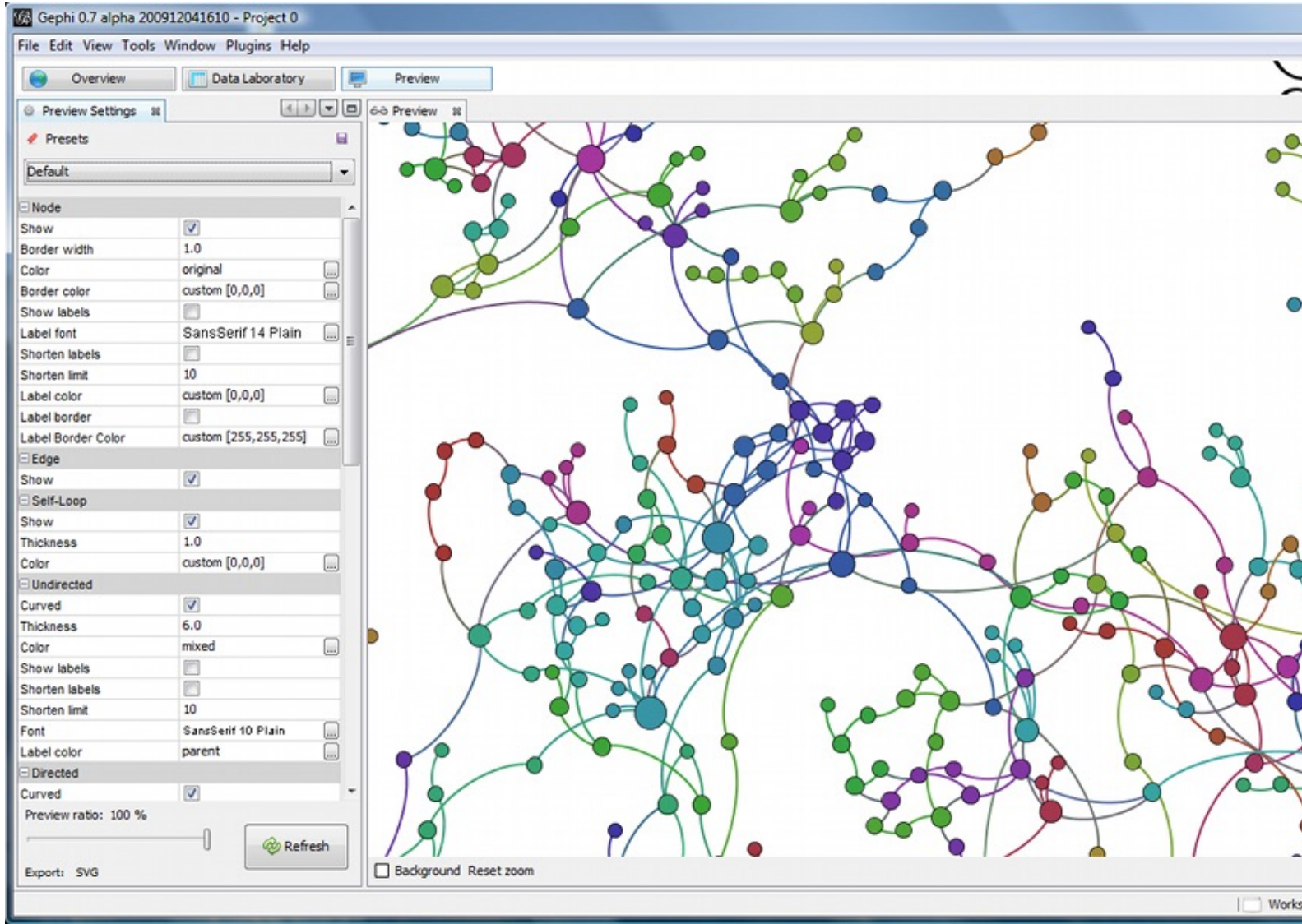


Gephi

Textual and spatial encoding

- LOD
- Relational data/graphs
- Various downloadable formats

Online version: https://gephi.org/gephi-lite/



NetworkX

Network Analysis in Python

[Documentation]

```
[8]  nx.degree(G)
```

```
DegreeView({'Aemon': 5, 'Grenn': 4, 'Samwell': 15, 'Aerys': 4, 'Jaime': 24, 'Robert'
```

```
[9]  degrees = dict(nx.degree(G))
     nx.set_node_attributes(G, name='degree', values=degrees)
```

```
[13] degree_df = pd.DataFrame(G.nodes(data='degree'), columns=['node', 'degree'])
     degree_df = degree_df.sort_values(by='degree', ascending=False)
     degree_df.head()
```

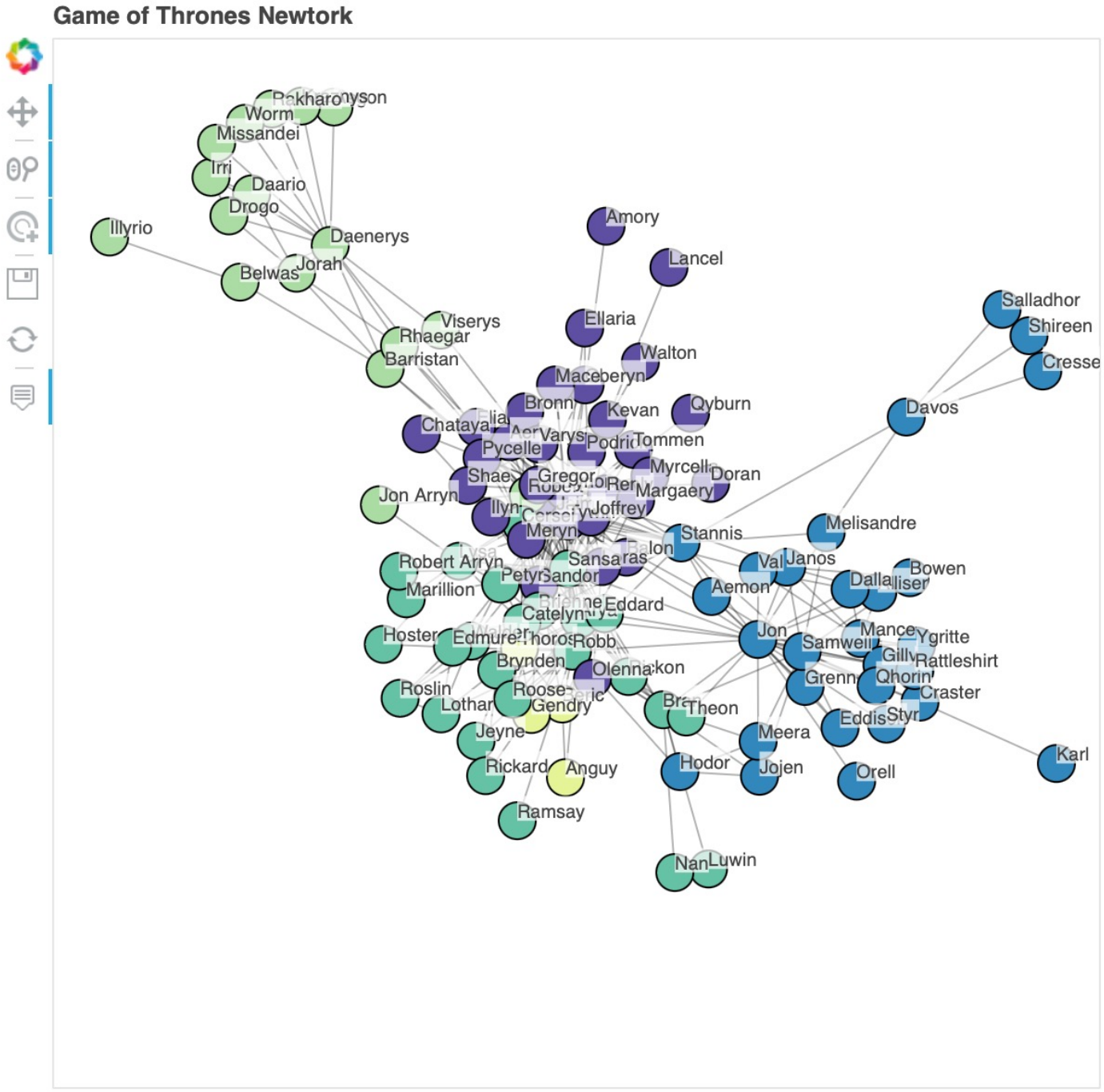
	node	degree
6	Tyrion	36
21	Jon	26
50	Sansa	26
48	Robb	25
4	Jaime	24

- Create **graphs** from scratch or from pre-existing data
- Run **analyses** on them, e.g.:
  - How many connections?
  - How strong are the connections?
- Basic **visualisation** methods

bokeh

http://bokeh.org

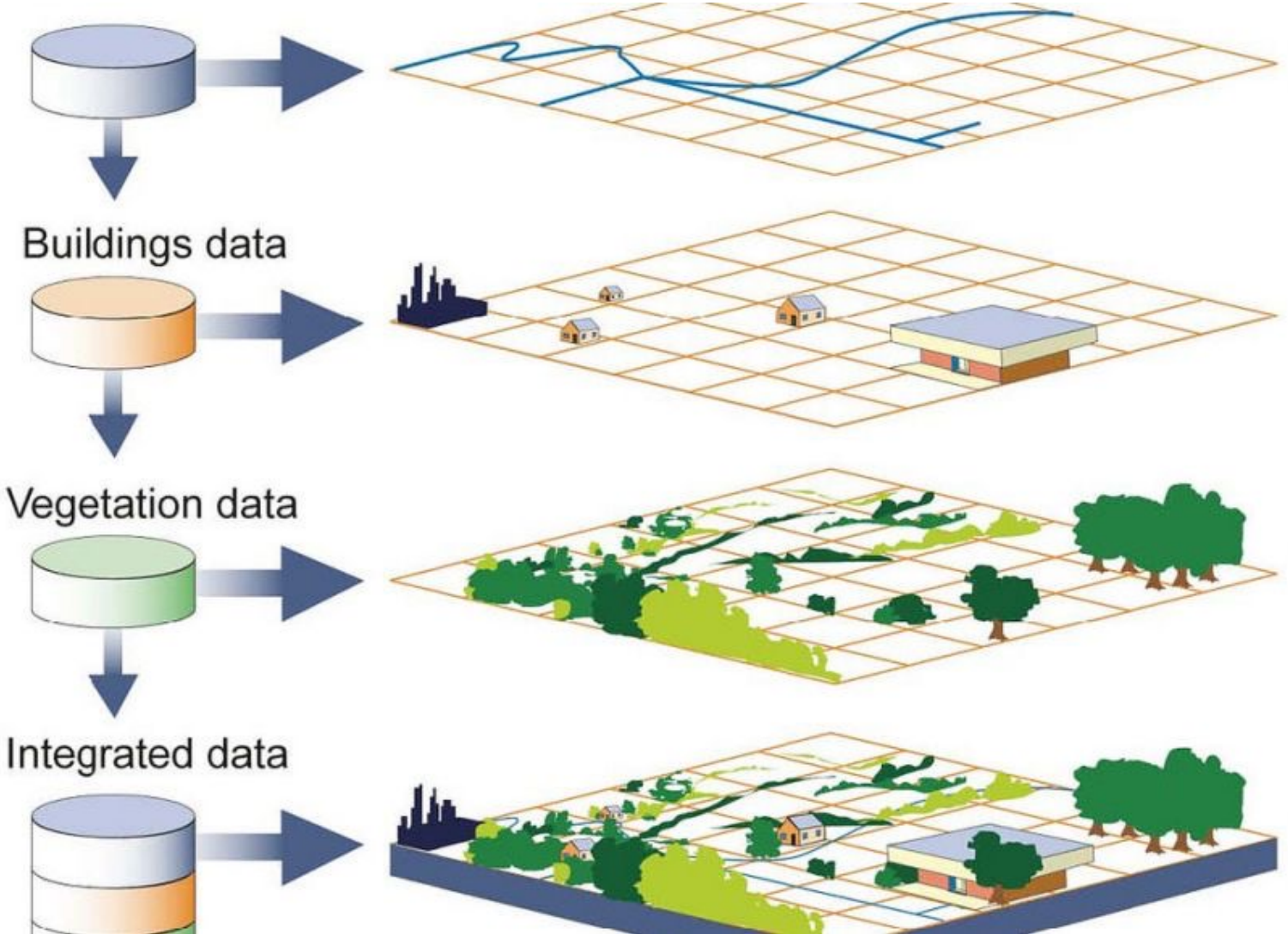
- **Python** library
- Create **interactive** visualisations
- Use these visualisations in **web pages**
- See an example here (based on *Game of Thrones*)



GIS

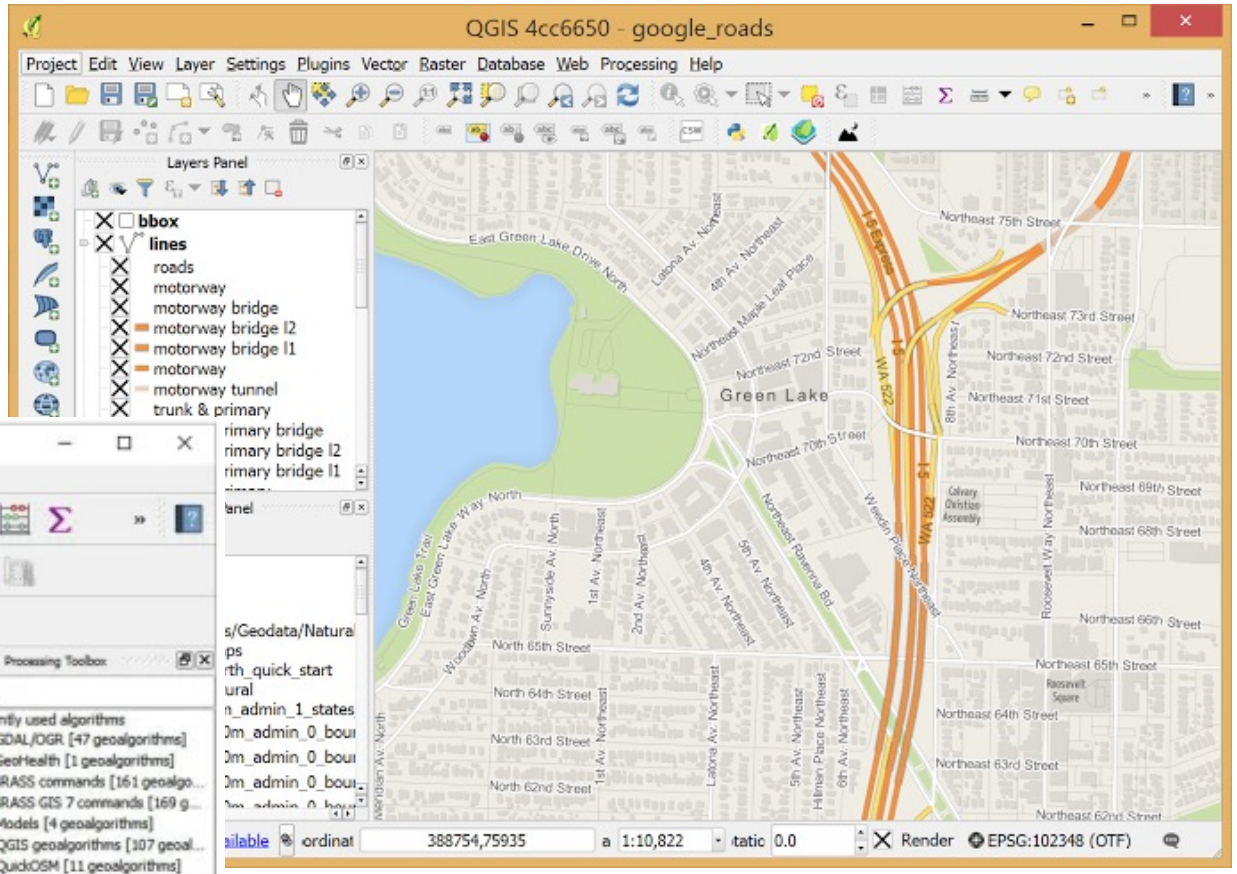
GIS = Geographic Information System(s)

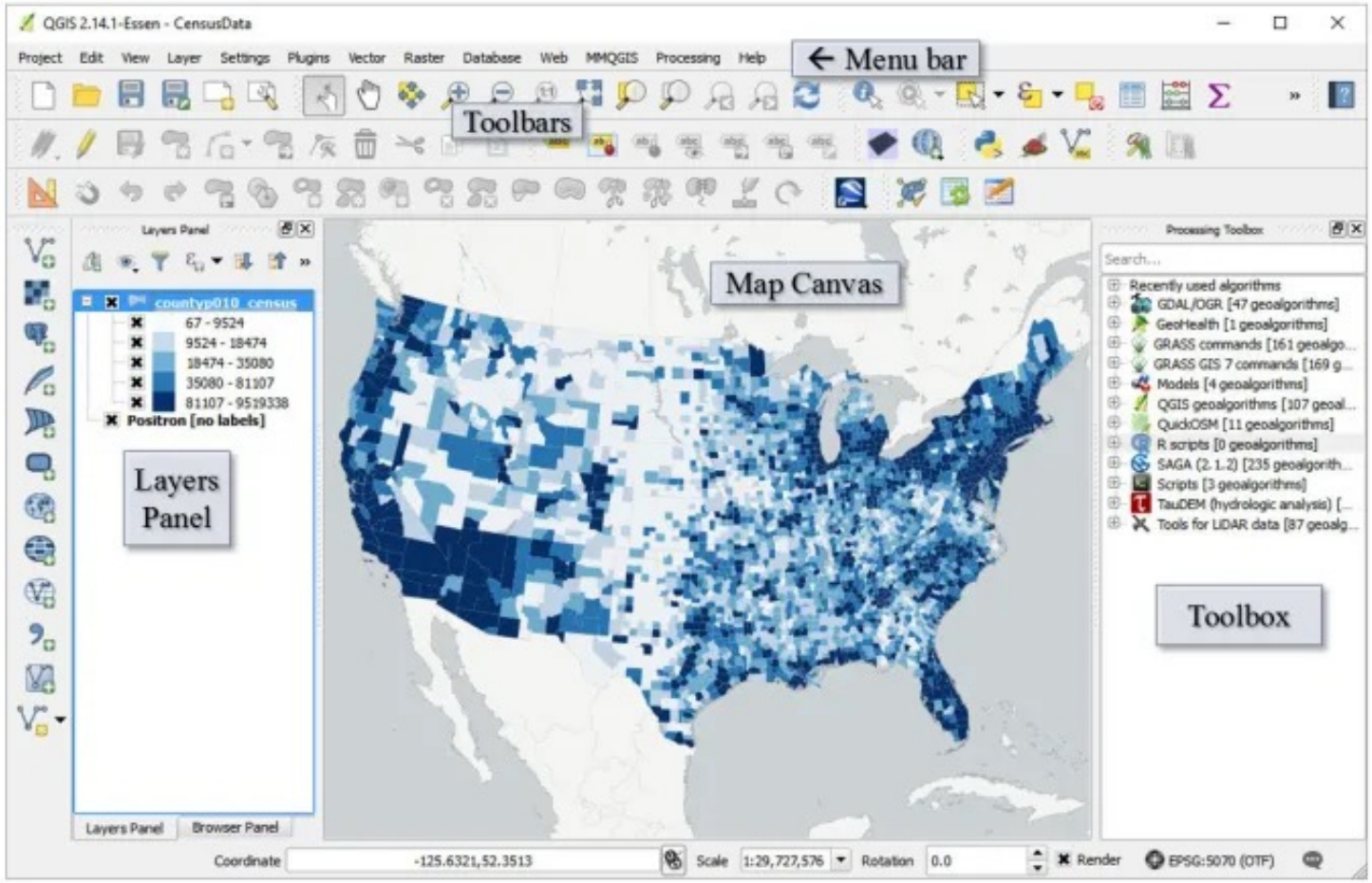
- computer-based tools used to store, visualize, analyze, and interpret geographic data



https://education.nationalgeographic.org/resource/geographic-information-system-gis/

QGIS

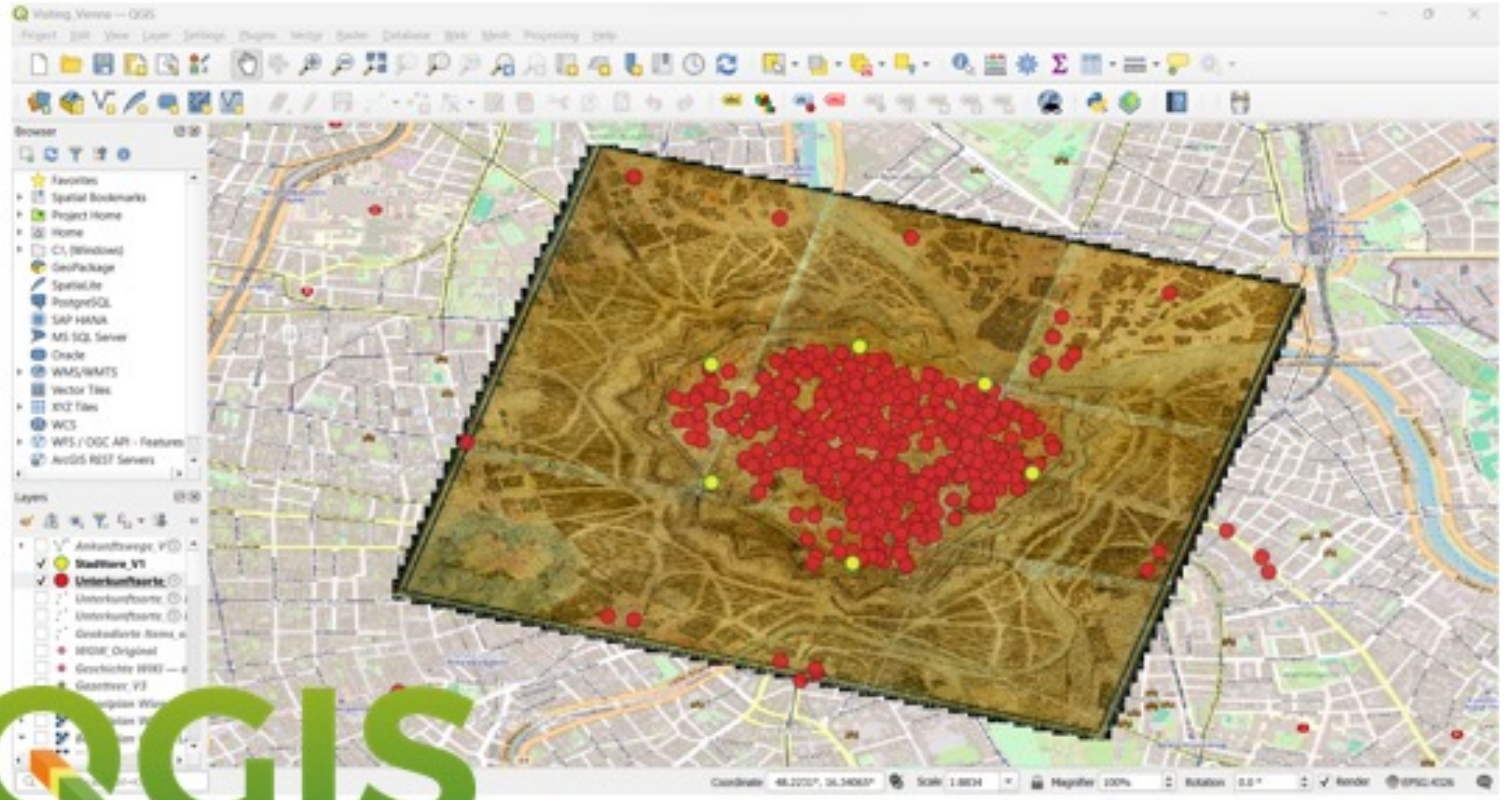




https://qgis.org/de/site/

Open-Source GIS software (spatial analysis, map visualisations, …)

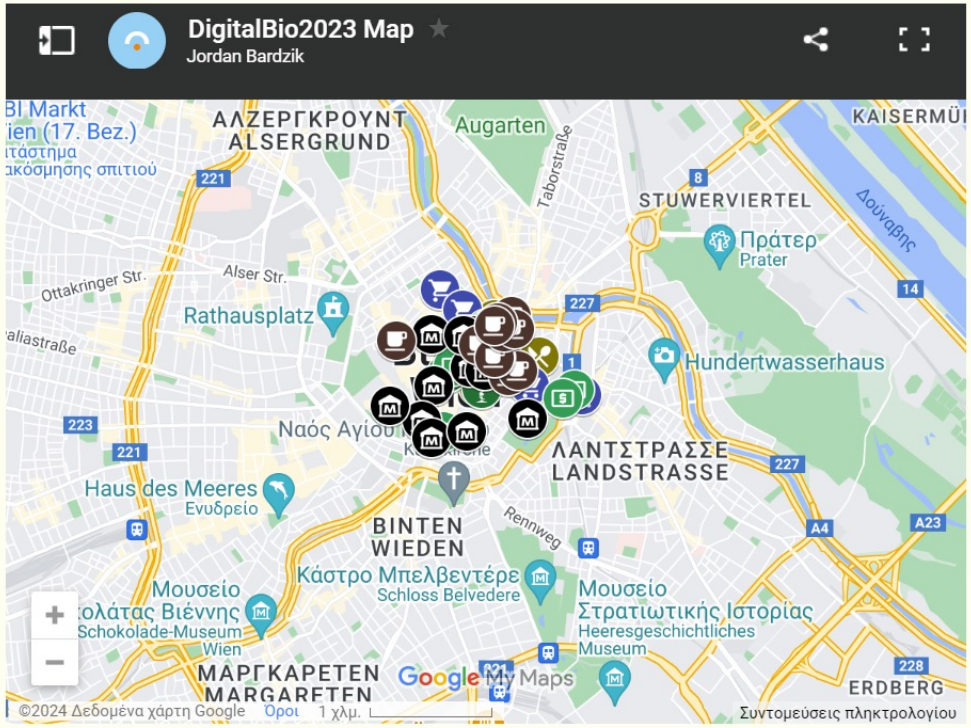
Exemplary Use Case



- ACDH-CH Project "Visiting Vienna"
- Mapping of arriving persons in early 18th century Vienna based on arrival list data
- Use of georeferenced historic plans of Vienna

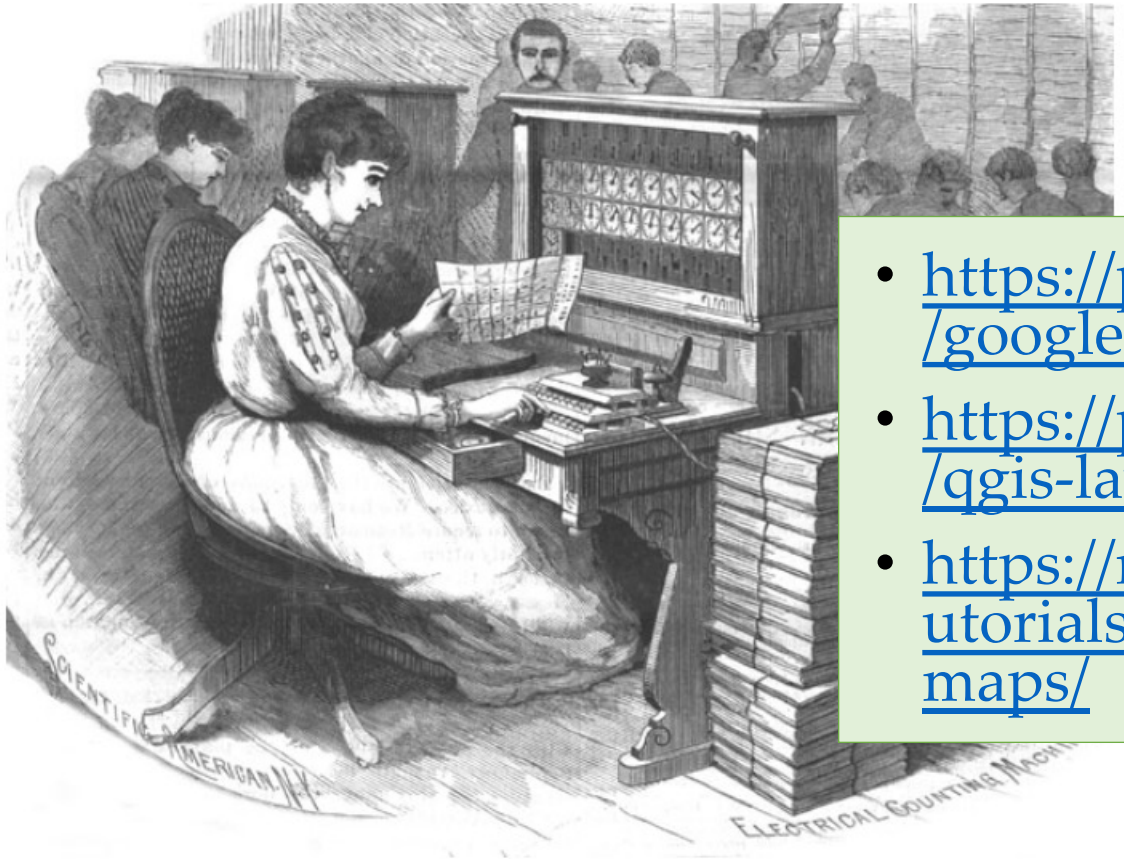
Google Maps

KML



https://www.google.com/maps/d/edit?mid=1o1p_2DPFs62j6OXx00ShFAvqg1JCIQw&usp=sharing

Tutorials

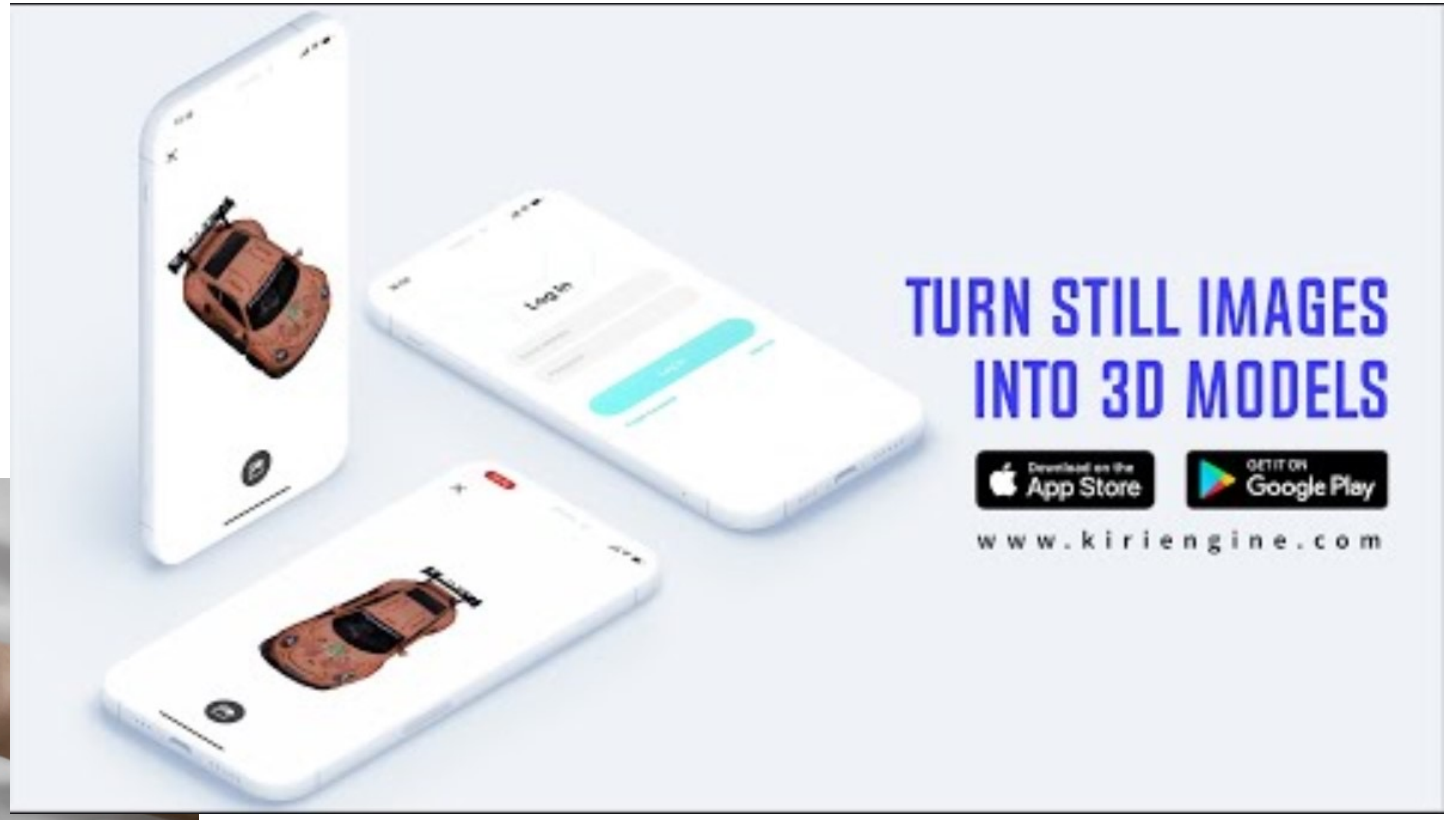


- https://programminghistorian.org/en/lessons/googlemaps-googleearth
- https://programminghistorian.org/en/lessons/qgis-layers
- https://multimedia.journalism.berkeley.edu/tutorials/create-custom-maps-in-google-maps/

3D Visualisation

KIRI Engine

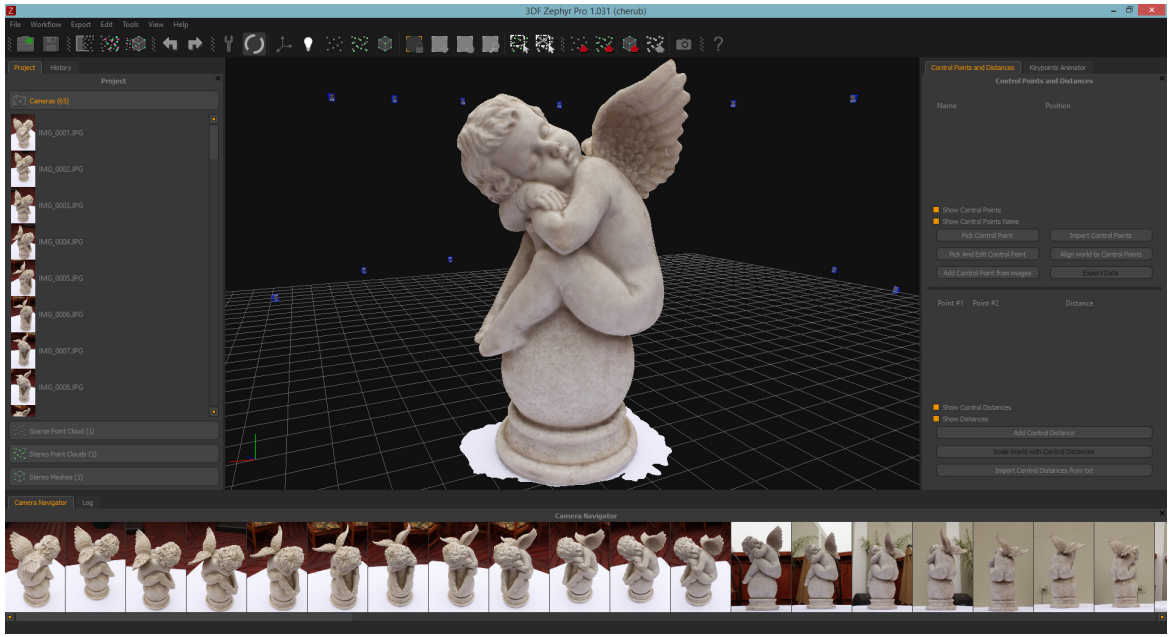
https://www.kiriengine.app/

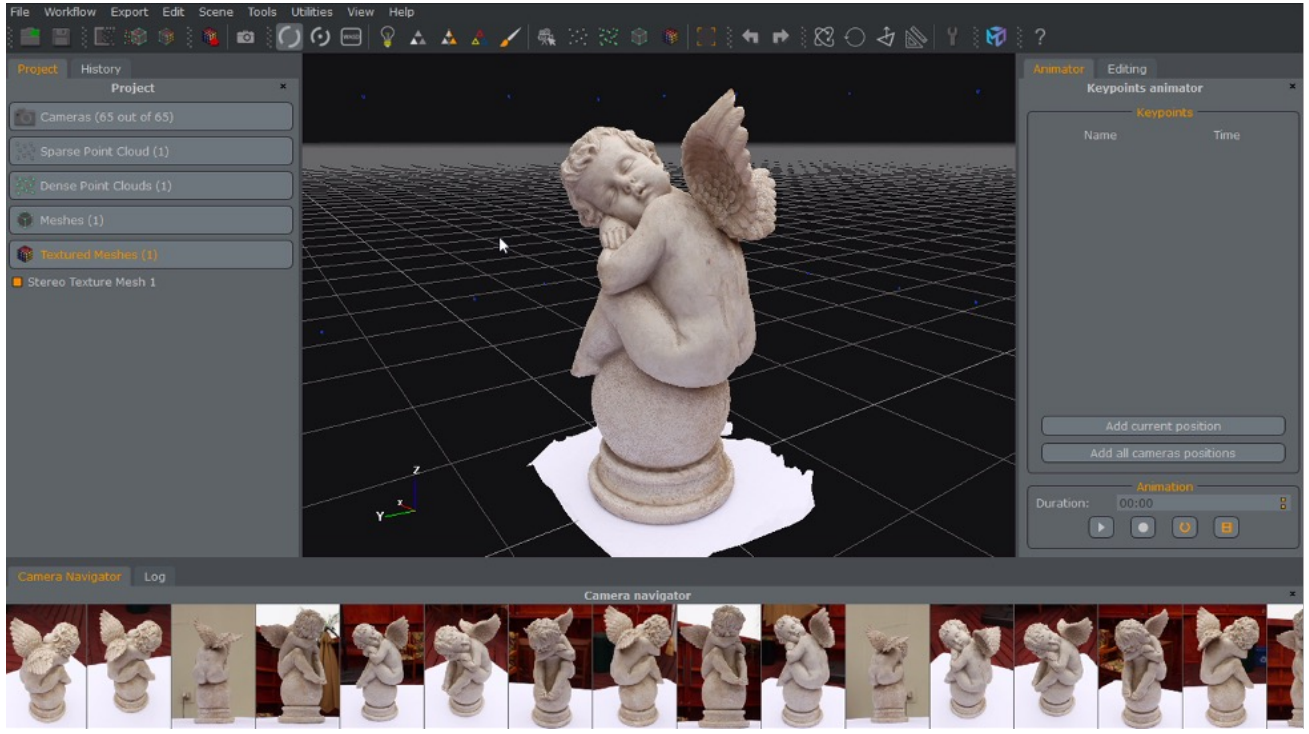


3DF **ZEPHYR**

The Complete Photogrammetry Solution

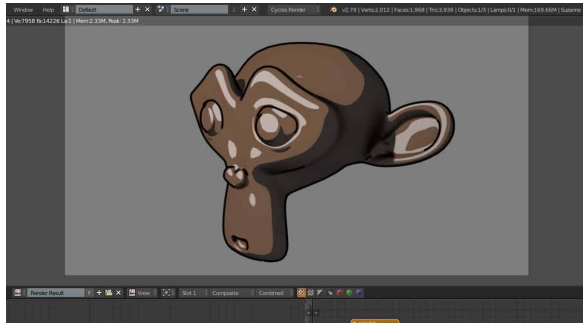
https://www.3dflow.net/3df-zephyr-photogrammetry-software/

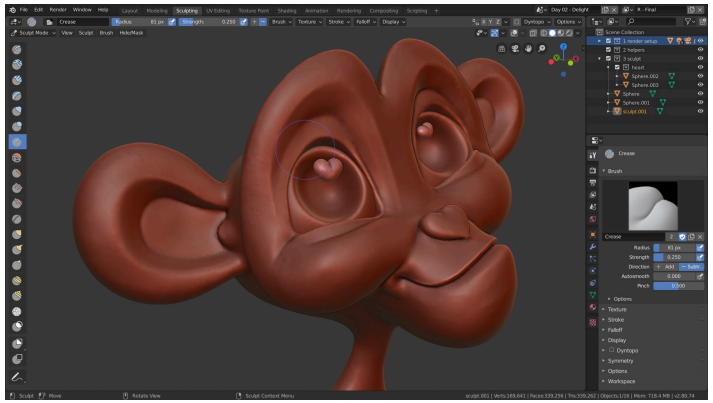


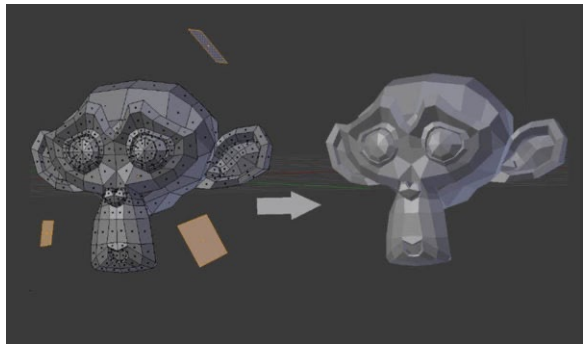


Free photogrammetry tool
(3D representations)

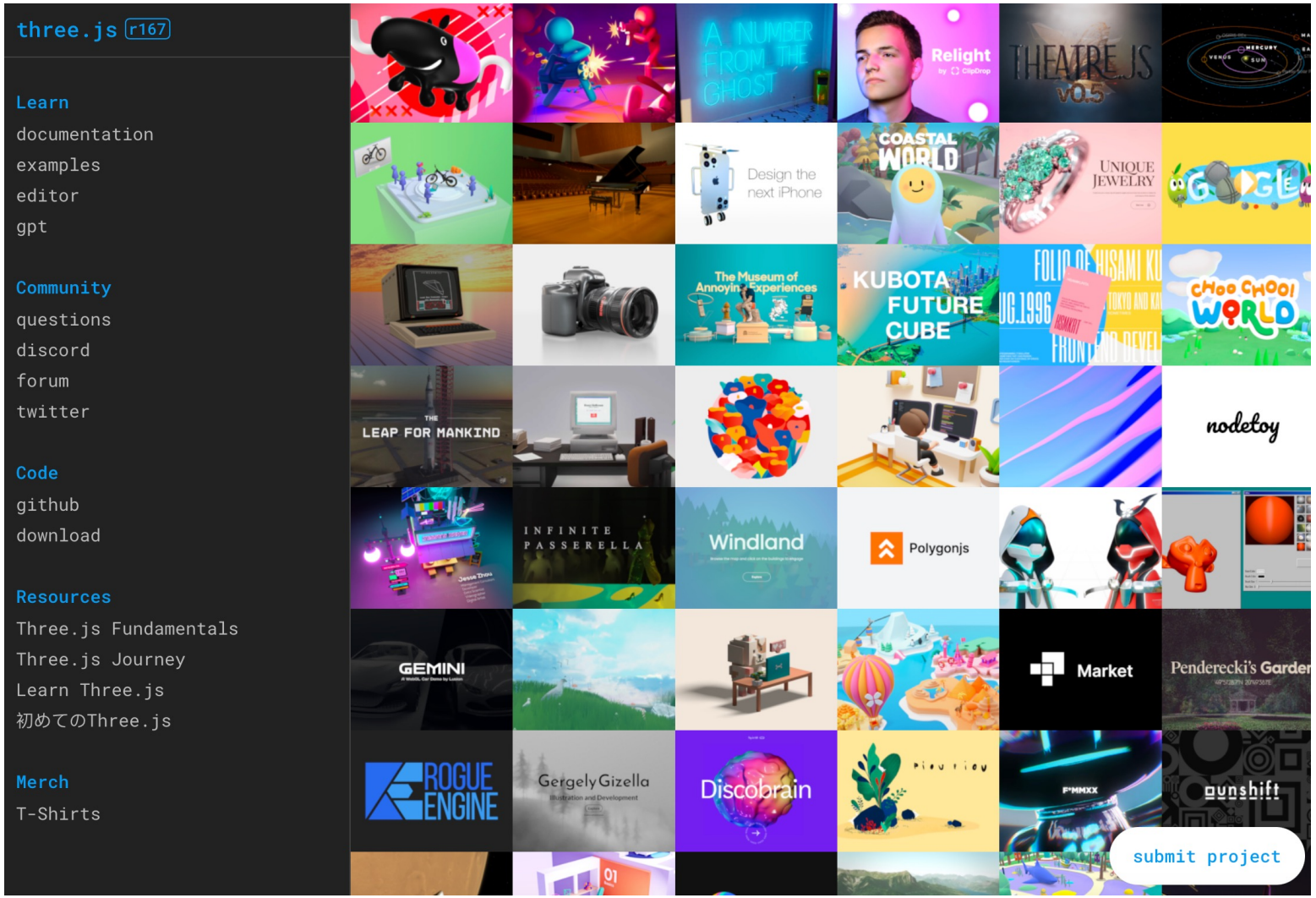
https://www.blender.org/







Free and Open source 3D creation suite.
It supports the entirety of the 3D pipeline (e.g. modeling, rigging, animation, simulation, rendering, compositing and motion tracking, even video editing and game creation.)



Three.js

https://threejs.org

JavaScript library

Semantic Web

2001

Tim Berners-Lee, James Hendler, & Ora Lassila, The Semantic Web, *Scientific American* vol. 285 issue 5, May 2001.

1999

I have a dream for the Web [...] Machines become capable of analyzing all the data on the Web [...] A "Semantic Web," which should make this possible, has yet to emerge, but when it does, the day-to-day mechanisms of trade, bureaucracy and our daily lives will be handled by machines talking to machines

Tim Berners-Lee with Mark Fischetti, Weaving the Web: The Original Design and Ultimate Destiny of the World Wide Web by Its Inventor, New York: Harper 1999, pp. 157-8.

https://en.wikipedia.org/wiki/Vienna

https://www.wikidata.org/wiki/Q1741

WIKIPEDIA
The Free Encyclopedia

Create account Log in

Vienna

226 languages

Article Talk Read View source View history Tools

From Wikipedia, the free encyclopedia

Coordinates: 48°12′30″N 16°22′21″E

This article is about the capital of Austria. For other uses, see Vienna (disambiguation).
"Wien" redirects here. For other uses, see Wien (disambiguation).
Not to be confused with Vienne (disambiguation).

This article **contains weasel words: vague phrasing that often accompanies biased or unverifiable information**. Such statements should be clarified or removed. *(September 2023)*

Vienna (/viˈɛnə/ *vee-EN-ə*;[9][10] German: *Wien* [viːn]; Austro-Bavarian: *Wean* [veɐ̯n]) is the capital, largest city, and one of nine federal states of Austria. Vienna is Austria's most populous city and its primate city, with about two million inhabitants[3][11] (2.9 million within the metropolitan area,[12] nearly one-third of the country's population), and its cultural, economic, and political center. It is the fifth-largest city proper by population in the European Union and the largest of all cities on the Danube river by population.

The city lies on the eastern edge of the Vienna Woods (*Wienerwald*)—the northeasternmost foothills of the Alps, that separate Vienna from the more western parts of Austria—at

Vienna
Wien (German)
Wean (Bavarian)

Capital city, federal state and municipality

Vienna (Q1741)

capital of and state in Austria
Wien | Vienna, Austria | W

In more languages
Configure

Language	Label	Description	Also known as
English	Vienna	capital of and state in Austria	Wien Vienna, Austria W
German	Wien	Bundeshauptstadt der Republik Österreich	AT-9 AT VIE W
Bavarian	Wean	Hauptstod vo Österreich	
French	Vienne	capitale de l'Autriche	Wien

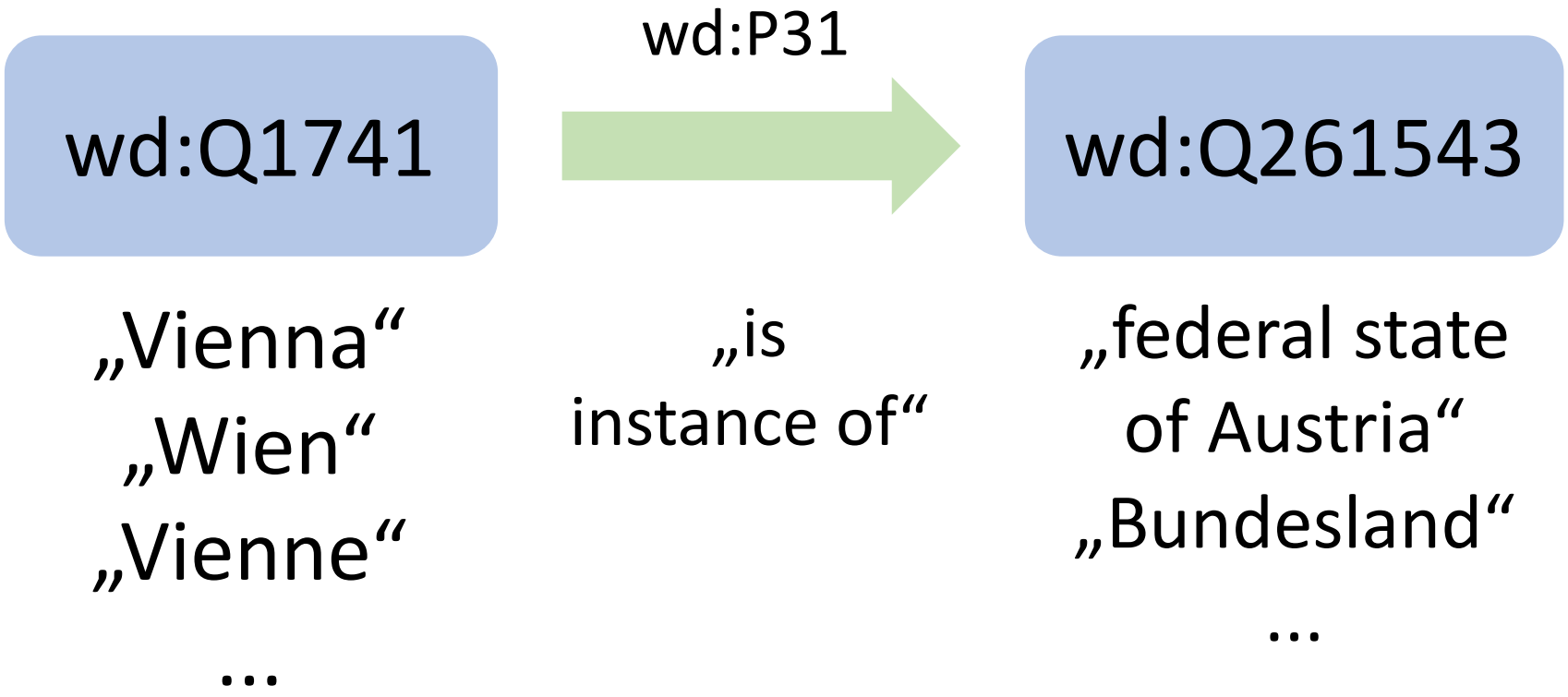
All entered languages

Statements

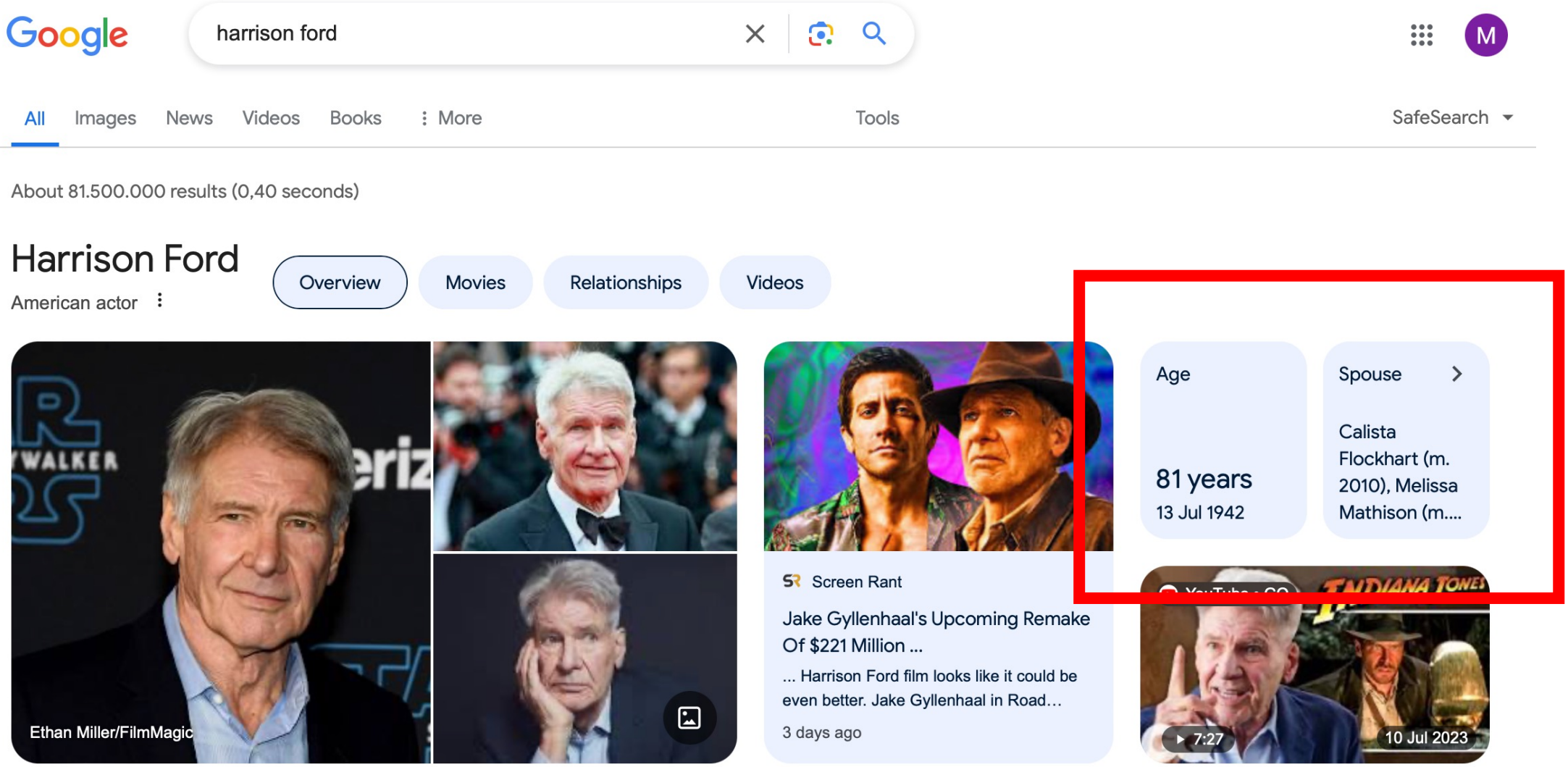
instance of	federal capital	0 references
	statutory city of Austria	0 references
	federal state of Austria	start time: 18 November 1920 Gregorian

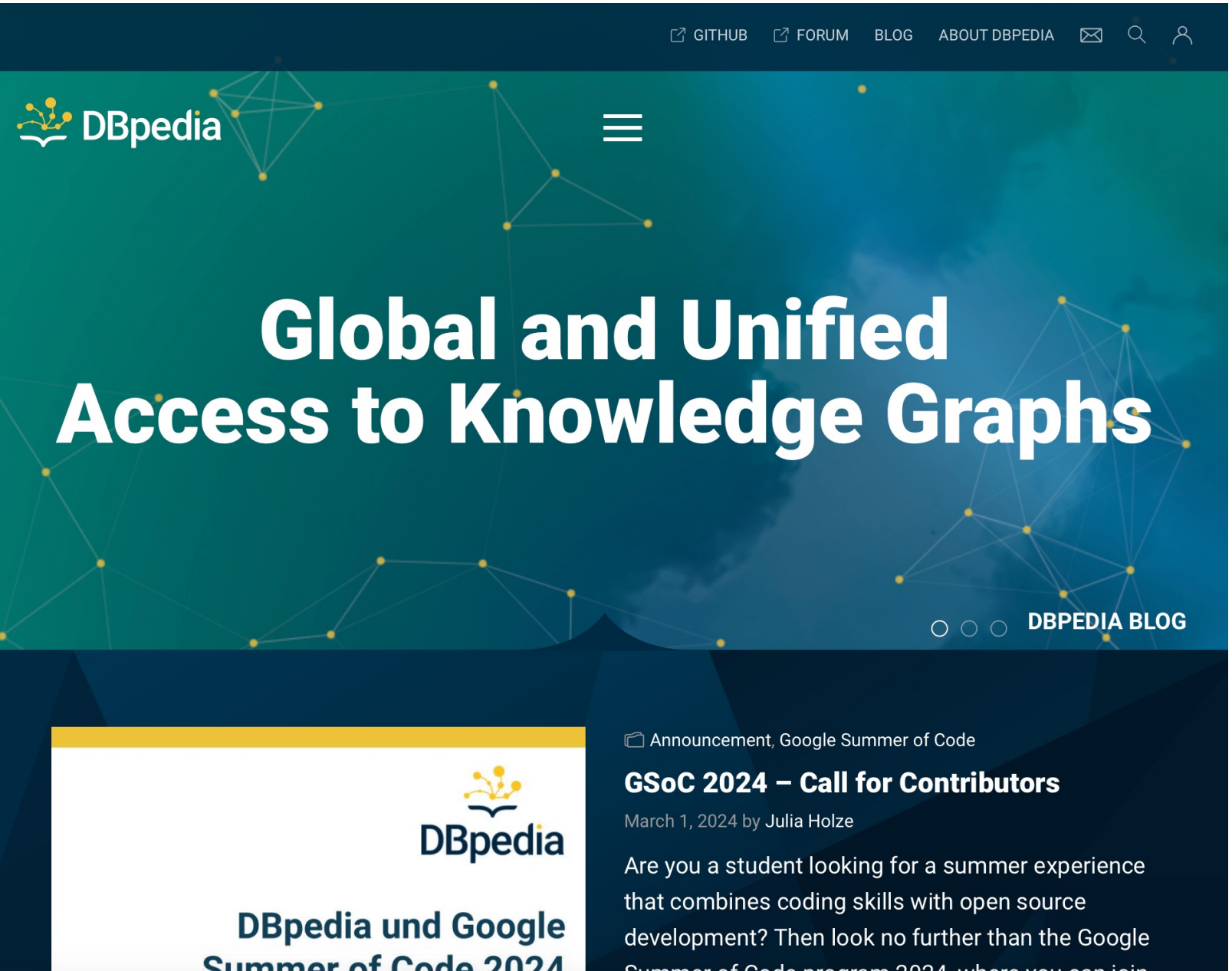
How does the Semantic Web work?

Vienna (/viˈɛnə/ ⓘ *vee-EN-ə*;[9][10] German: *Wien* [viːn] ⓘ; Austro-Bavarian: *Wean* [veɐ̯n]) is the capital, largest city, and one of nine federal states of Austria. Vienna is Austria's most populous city and its primate city,



Google Knowledge Panel





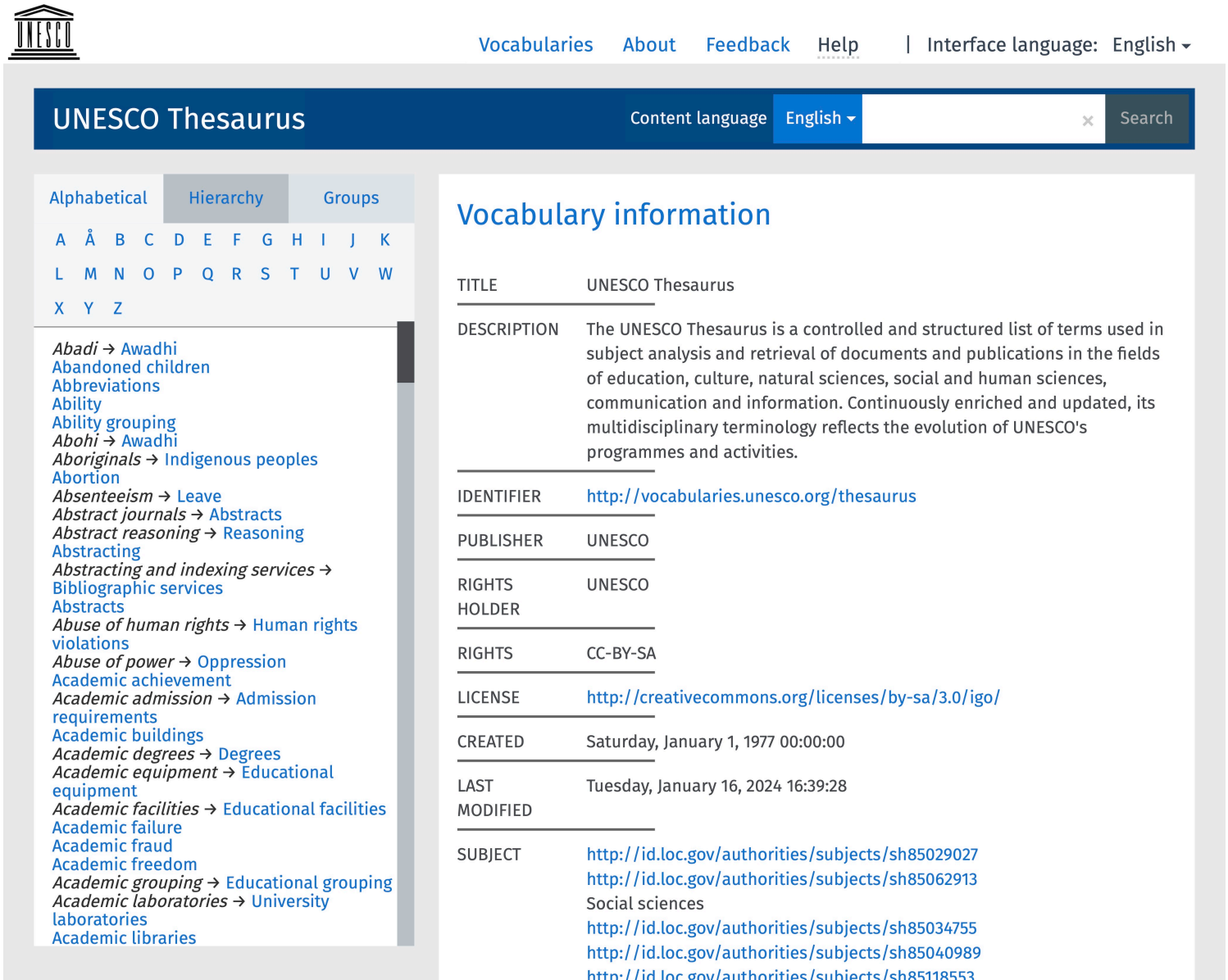
DBpedia

[Faceted browser]

[Query interface]
(advanced)

https://dbpedia.org/page/Vienna

Controlled vocabularies



- Define **terms** that should be used in a domain / for a task
- Allow all applications (or users) to understand what we are talking about **unequivocally**!
- Usually based on the SKOS model
- Some examples include the UNESCO Thesaurus (on the left)

Some ways to go...

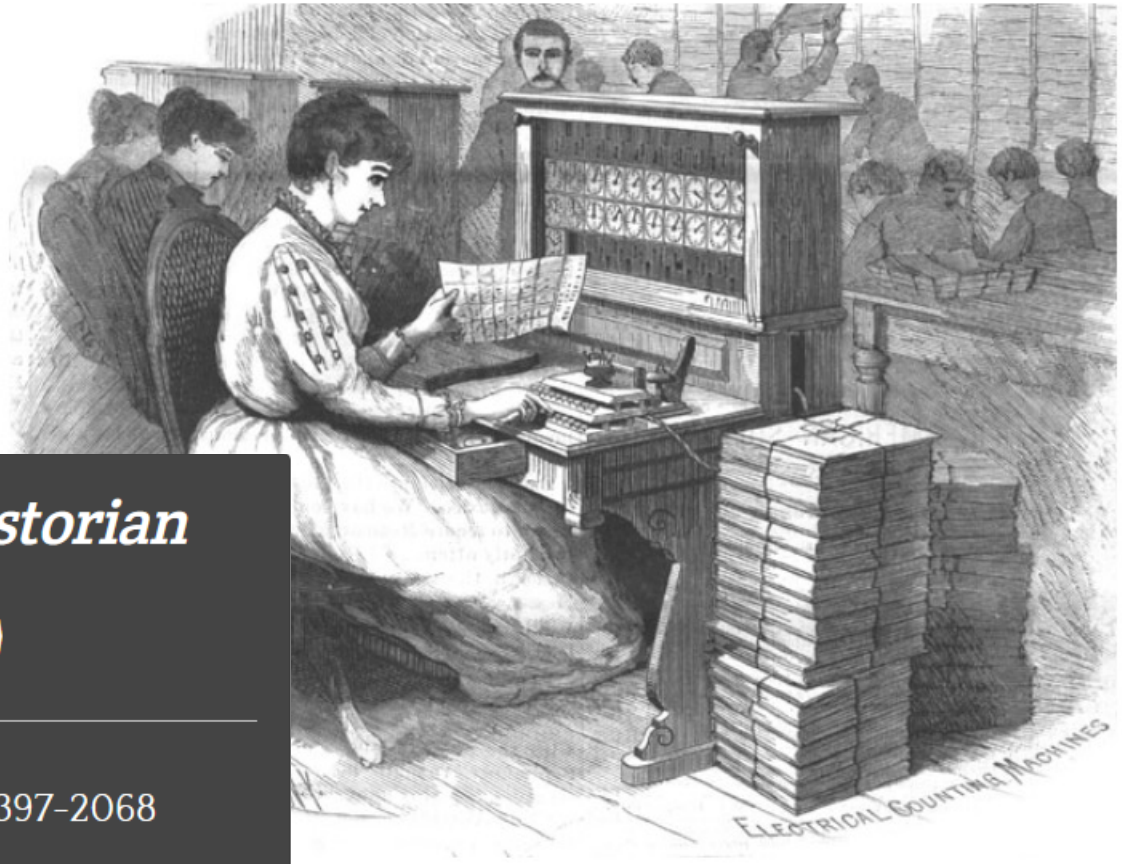
- Search for any Wikipedia page that might interest you, and click on **„Tools > Wikidata item“**... you can compare the structured data to the original page or browse through the entities by clicking on the links
- (More advanced) You can load a sample Neo4j graph database by using the Neo4j Sandbox, for example a database on „Movies“. Neo4j is a graph database for so-called „labeled property graphs“
- Have a look at the ACDH services, ARCHE and Vocabs, which provide data according to Semantic Web recommendations

Further resources to educate yourself

Programming Historian

https://programminghistorian.org/

(in English, French, Portuguese, Spanish)



[PH]

Enter *Programming Historian*

(The initial English version)

101 lessons

ISSN: 2397-2068

We publish novice-friendly, peer-reviewed tutorials that help humanists learn a wide range of digital tools, techniques, and workflows to facilitate research and teaching.

DARIAH-CAMPUS

https://campus.dariah.eu/

Research infrastructures 38 | Data management 37 | eHeritage 27 | DH 24 | Lexicography 22 | Open science 22 | Training and education 22 | Digital Archives 18

Data visualisation 16 | Open access 14 | Scholarly editions 11 | Metadata 10 | Repositories & Collections 10 | Big data 9 | Scholarly practice 9

Scholarly publishing 7 | Citizen science 6 | Computational Imaging 6 | EOSC 6 | Multimodality 6 | Data modeling 5 | Spatial humanities 5 | Sustainability 5

TEI 5 | XML 5 | Design Thinking 4 | Feminism 4 | Geotagging 4 | Open education 4 | Project management 4 | Controlled Vocabularies 3 | Source Criticism 3

Augmented reality 2 | Digital Source Criticism 2 | History of Technology 2 | Information Architecture 2 | Ontologies 2 | Semantic Web 2 | Software development 2

Sound studies 2 | Editing tools 1 | Game Studies 1 | Literacies 1 | Machine Learning 1 | Maker Culture 1 | MOOC 1 | SKOS 1 | Speech technology 1 | XPath 1

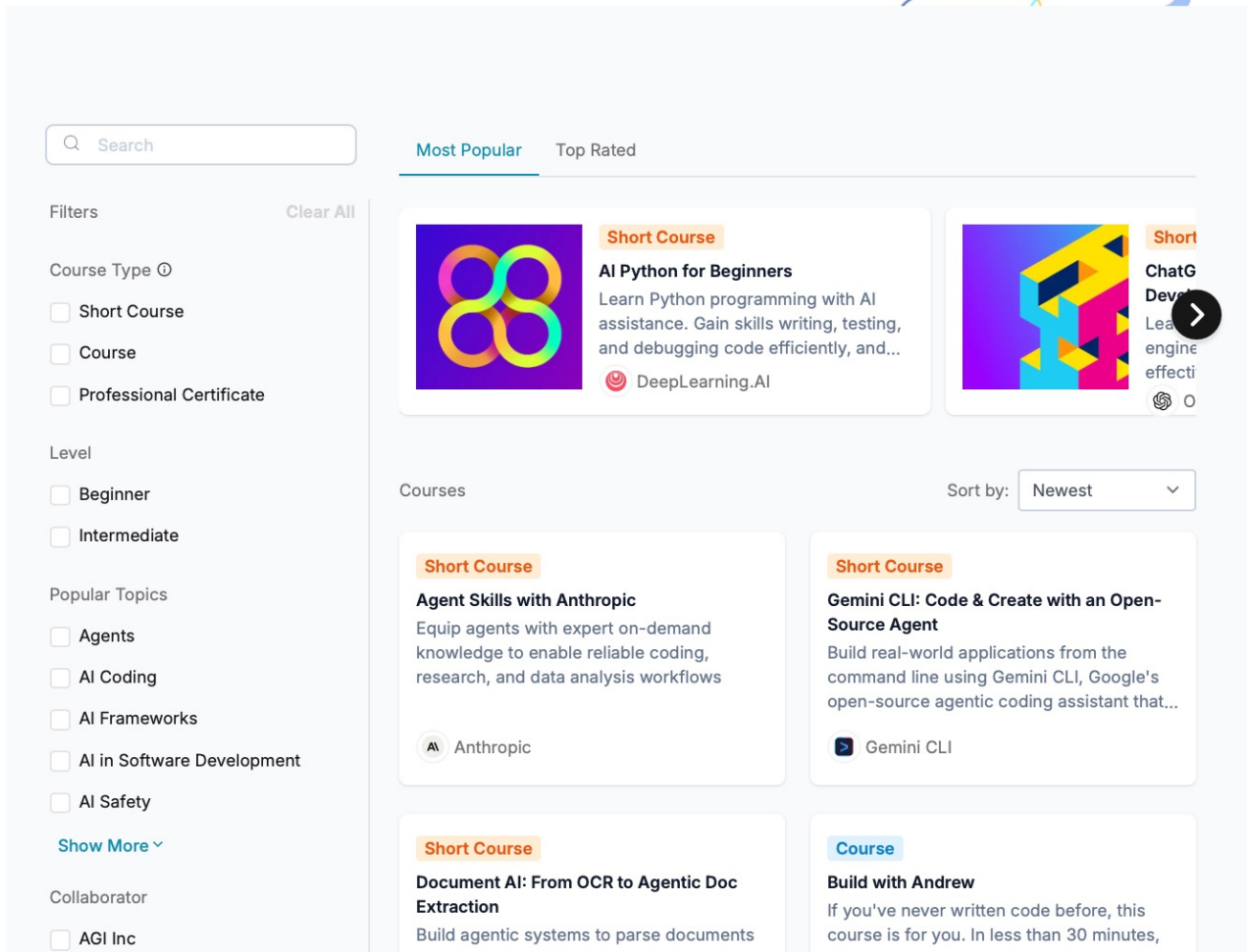
XSLT 1

Looking for learning resources?

DARIAH-CAMPUS is a discovery framework and hosting platform for learning resources.

DeepLearning.AI

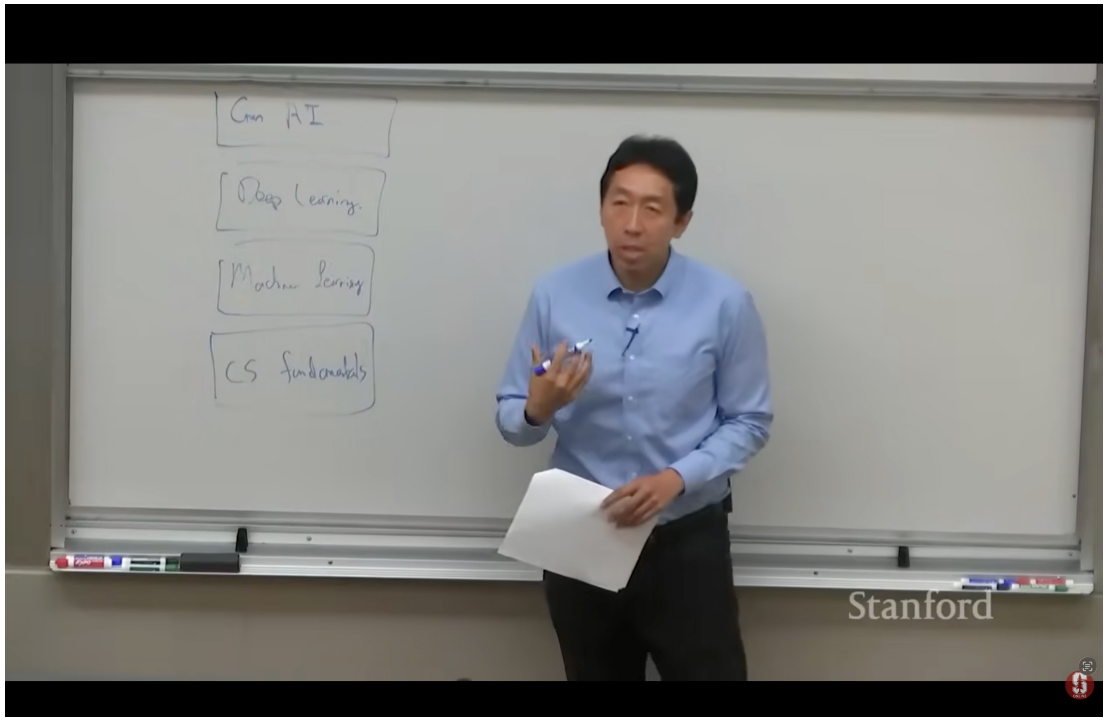
https://www.deeplearning.ai/courses/



Stanford | ENGINEERING

Stanford CS230 | Autumn 2025 on Deep Learning

https://youtu.be/_NLHFoVNlbg



TOOL GALLERIES

ACDH Tool Gallery 12.1

Generative KI & Vibecoding

https://www.oeaw.ac.at/acdh/newsevents/event-series/acdh-ch-tool-galleries

The ACDH Tool Galleries, which take place **three times a year**, are workshops dedicated to providing information and **hands-on training in the use of specific digital humanities tools**. Participants are given the opportunity to engage with the tools and to learn from experts with first hand experience in the application of digital methods in the field of humanities research. They are a service **open to everyone** but directed, in particular, at humanities scholars who would like to learn about and engage with computer-based methods in order to implement them in their own research.

Summer Schools

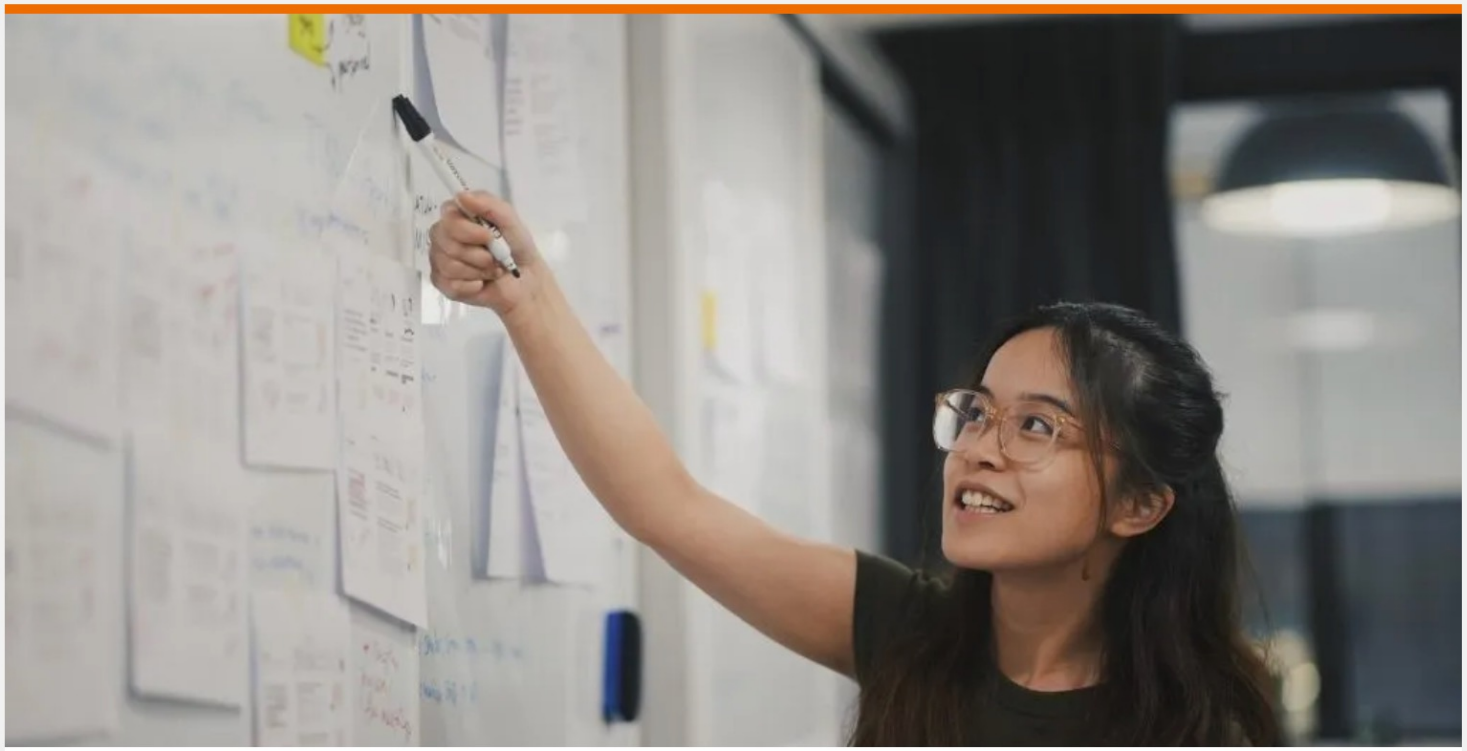
- European Summer University in Digital Humanities „Culture & Technology“
  - https://digihubb.centre.ubbcluj.ro/
- Edirom Summer School
  - https://ess.uni-paderborn.de/
- Helsinki Digital Humanities Hackathon (DHH)
  - https://www.helsinki.fi/en/digital-humanities/helsinki-digital-humanities-hackathon-2023-dhh23
- Digital Humanities @Oxford Summer School (DHOX)
  - https://digital.humanities.ox.ac.uk/digital-humanities-oxford-summer-school

→ (application for) financial support possible in most cases

CLARIAH-AT

[Funding for early career researchers]

- Transkribus
- Software licenses
- Research trips



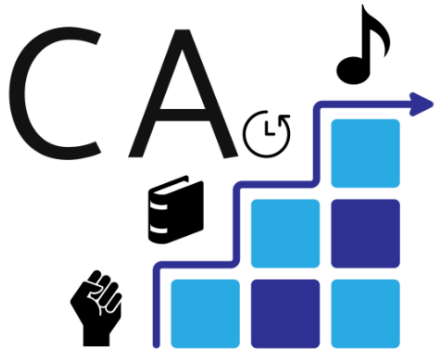
Early Career Förderungen

CLARIAH-AT hat sich zum Ziel gesetzt, Junior Researchers, die im Bereich der Digital Humanities forschen und nur geringe oder keine anderen Fördermöglichkeiten haben, finanziell zu unterstützen. Die Förderungen stellen eine Investition in die Zukunft und Weiterentwicklung der Digital Humanities in Österreich dar. Derzeit können folgende Förderungen beantragt werden:

Introduction to Cultural Analytics & Python

Designed by Melanie Walsh // Powered by Jupyter Book

This website hosts an online textbook, *Introduction to Cultural Analytics & Python*, which offers an introduction to the programming language Python that is specifically designed for people interested in the humanities and social sciences.

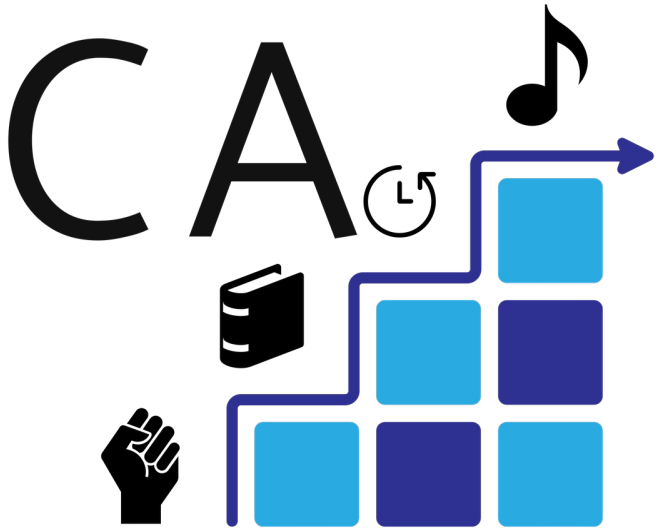


This book demonstrates how Python can be used to study cultural materials such as song lyrics, short stories, newspaper articles, tweets, Reddit posts, and film screenplays. It also introduces computational methods such as web scraping, APIs, topic modeling, Named Entity Recognition (NER), network analysis, and mapping.

These materials were originally created to support "Introduction to Cultural Analytics: Data, Computation & Culture," an undergraduate course taught at Cornell University and the University of Washington.

What is Cultural Analytics?

Cultural analytics is the study of culture with computational methods. Culture is a broad term that includes



https://melaniewalsh.github.io/Intro-Cultural-Analytics/welcome.html

Other tool collections

forTEXT

Literatur digital erforschen

Routinen ▾ Ressourcen ▾ Tools ▾

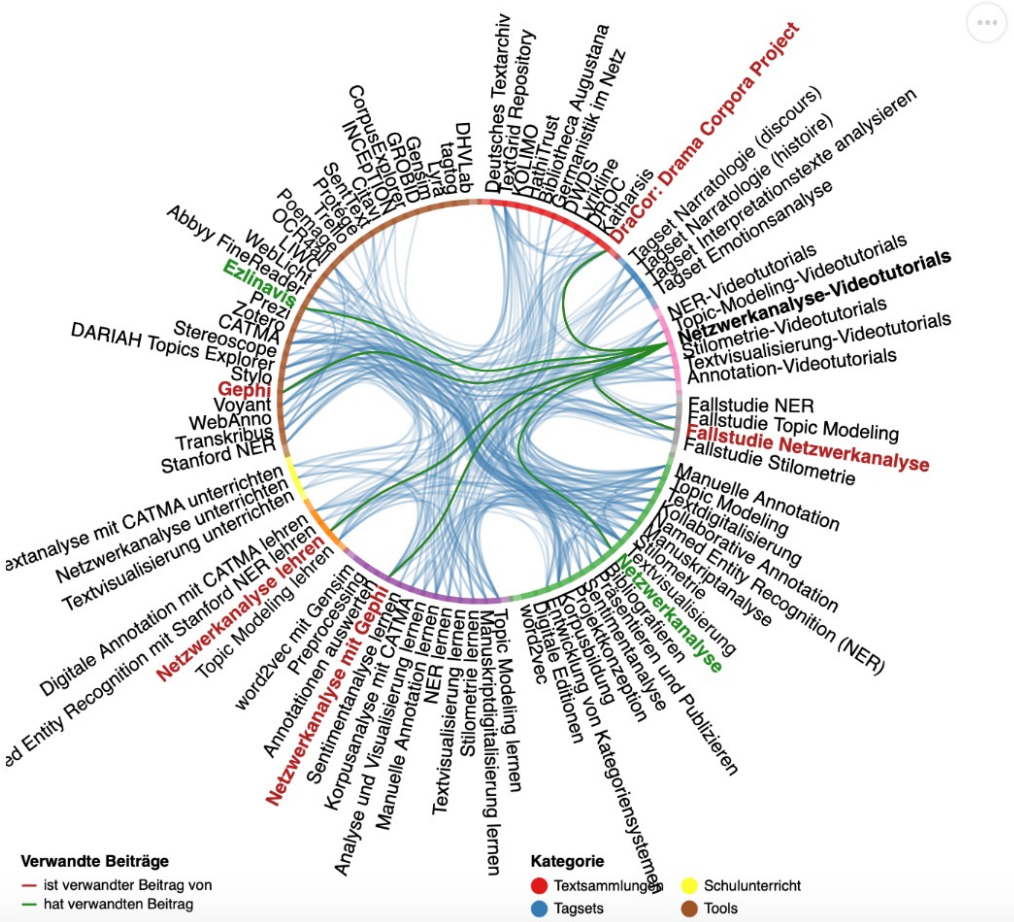
https://fortext.net

Digitale Textanalyse – leicht gemacht

Wir bieten einsteigerfreundlich aufbereitete zitierfähige Methodenbeschreibungen, Textsammlungen und Tools – von Digitalisierung über Annotation zu Interpretation und Visualisierung von Literatur. Außerdem finden Sie hier Materialien zum Selbstlernen und zum Lehren an Uni und Schule.

Die forTEXT-Elemente

In forTEXT werden Ihnen verschiedene Routinen, Ressourcen und Tools angeboten, bewertet und – zum Teil mit der Hilfe von Videos – erklärt. Die folgende Visualisierung ermöglicht Ihnen, unsere Inhalte zu entdecken.



Neuigkeiten

Call for Papers 2024: Textannotation in der Hochschullehre OUT NOW!

Im Rahmen der forTEXT Hefte haben wir den ersten Call for Papers veröffentlichen. Wir freuen uns über Einreichungen zu Lehrkonzepten zum Thema "Textannotation in der Hochschullehre"!

Weiterlesen

Korpusbildung und Textvisualisierung jetzt als Themenhefte verfügbar!

Weiterlesen

1. CATMA Workshop 2024

Wir freuen uns auf den ersten CATMA Workshop in diesem Jahr mit euch!

Weiterlesen

Routinen

20. April 2023
word2vec mit Gensim

19. April 2023
word2vec

4. Juli 2022
Digitale Editionen

Ressourcen

3. Dezember 2020
DraCor: Drama Corpora Project

21. September 2020
Emotionsanalyse

27. Juli 2020
Tagset Interpretationstexte

Awesome Digital Humanities

Tools, resources, and services supporting the Digital Humanities.

View the Project on GitHub
dh-tech/awesome-digital-humanities

This project is maintained by dh-tech

Hosted on GitHub Pages — Theme by orderedlist

Awesome Digital Humanities

awesome

> *Software for humanities scholars using quantitative or computational methods.*

This is a curated list of tools, resources, and services supporting the Digital Humanities. Contributions are welcome!

Contents

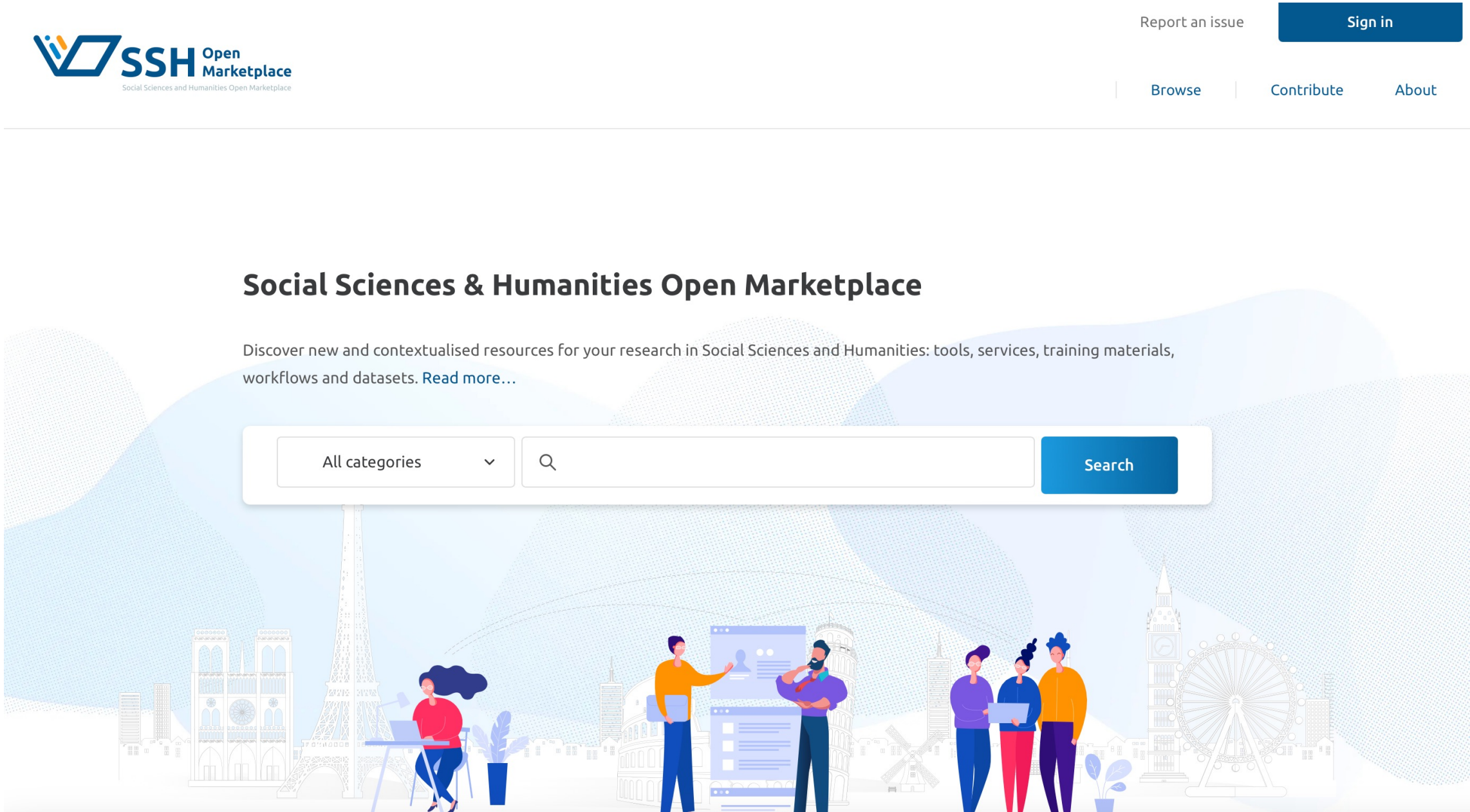
- Bibliography and Sources Management
- Corpus linguistics
- Data Collection
- Data Analysis
- Data Extraction and Conversion
- Data Annotation
- Data Augmentation
- DH Centers
- Document Management and Processing
- Journals
- Organizations and Research Infrastructures
- Other Resources
- Platforms
- Publishing
- Tool Building and Rapid Prototyping
- Twitter
- User Guides and Training Materials
- Visualization

Bibliography and Sources Management

https://dh-tech.github.io/awesome-digital-humanities/

+ More awesome lists...

SSH Open Marketplace: https://marketplace.sshopencloud.eu



Example of a **workflow** in SSHOMP

https://marketplace.sshopencloud.eu/workflow/4qFarh

Home / Workflows / Create a dictionary in TEI

Copy to clipboard

Create a dictionary in TEI

This scenario sets out the best practices for creating a born-digital dictionary, especially with the TEI (Text Encoding Initiative). However, buiding a standardized lexicographical dataset is not only a data format problem, it is also an intellectual and technical process where one has to choose how to model their data, and with which tools operate in order to create an easy-to-use and sustainable resource.

Details

ACCESS

License Creative Commons Attribution 4.0 International

CATEGORISATION

Activity Encoding

Keyword recommended

Discipline Linguistics and Literature

Staying up to date

Staying up to date

Websites

- DHd blog: https://dhd-blog.org/
- dha website: https://clariah.at/
- H-Net (Humanities and Social Sciences Online): https://networks.h-net.org/ (identify specific networks interesting to you)

Mailing lists

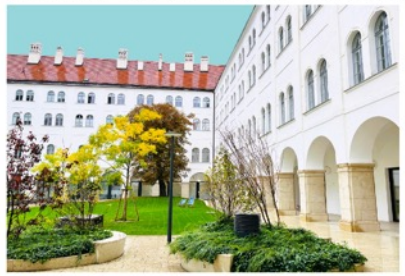
- Humanist: https://dhhumanist.org/membership_form.php
- DHd-Mailingliste: https://lists.uni-hamburg.de/mailman/listinfo/dhd
- Linguist: https://linguistlist.org/subscribe/
- Corpora: https://list.elra.info/mailman3/postorius/lists/corpora.list.elra.info/
- DH Summer Institute: https://lists.uvic.ca/mailman/listinfo/institute

ÖAW AUSTRIAN ACADEMY OF SCIENCES

ACDH – AUSTRIAN CENTRE FOR DIGITAL HUMANITIES

ACDH INTERNSHIPS

FEBRUARY 2027 AT THE ÖAW CAMPUS ACADEMY IN VIENNA



CC-BY 4.0 Maria Wiederländers

Are you a student looking for hands-on work experiences at an academic research institute?
Are you thinking of an academic career in the humanities?

Would you like to know how to apply digital methods and tools to make existing research more insightful and ask new research questions?

Then the internship at the **Austrian Centre for Digital Humanities** is for you!

WHO WE ARE

The ACDH is a leading research institution in digital humanities (DH) and part of the Austrian Academy of Sciences (ÖAW). We work closely with international research networks and support local research interests.

Currently, our teams work on various exciting projects, and we focus on many fields of expertise.
Some of them are:

- **DH Research & Infrastructure:** DH methods, tools, data interoperability, programming solutions, education, and outreach, a.o.
- **Linguistics:** language variation and language change, dialectology, corpus linguistics, lexicography, morphology, syntax, dialects in various languages, a.o.
- **Literary & Textual Studies:** corpora, digital editions, periodicals and newspapers, and correspondences, manuscripts and typescripts, a.o.
- **Musicology:** music as cultural heritage, editions, choir books, audiovisual research, a.o.

WHAT WE OFFER

During your time with us we want you to:

- gain hands-on work experience
- get insights into the workings of an academic research institute
- find out more about the focus on DH scholarship
- understand what a career as a DH researcher may entail

We offer internships of 80 hours (20 hrs a week for 4 weeks) or 120 hours (30 hrs a week for 4 weeks).

In addition to getting to know a project, we offer our **Meet the Researchers** sessions: researchers from different areas of expertise offer additional **workshop sessions** to expand your DH knowledge.

Also, depending on your university, you can be awarded **ECTS points** that count towards your degree.

WHO ARE WE LOOKING FOR

In general: students from DH and the fields of research that we work on!

In particular: applicants with backgrounds in **philology** (particularly **German philology**), **linguistics**, **literary studies**, **musicology, or other philological, cultural or sociological paradigms**. Some familiarity with **XML** and **web technologies** such as **HTML5**, **CSS** & **Javascript**, or programming languages like **Python**, is helpful, but not conclusive.

Regardless of your specialisation, we look for interns with: good **reasoning skills**, excellent **time management**, good **social skills**, **reliability** and **accuracy**. We also expect at least a basic adeptness in working with computers and digital methods. With regards to language skills, you should be able to communicate efficiently in either **German** or **English**, ideally both.

APPLY NOW!

Find out more on our website:

Application is open until
November 15, 2026

WWW.OEAW.AC.AT/ACDH

ÖAW

austrian centre for digital humanities

austrian centre for digital humanities

Apply for an internship at ACDH (February 2027):
https://www.oeaw.ac.at/acdh/knowledge-transfer/internships

Deadline: 15 November 2026

Explore!

Time plan for tool exploration

(1) Selection of a tool unfamiliar to you (5-10 min.)

(2) Exploration of tool (30 min.)

(3) a) Exploration of another new tool (15 min.)

b) Further exploration and discussion together with colleagues interested in the same (group of) tool(s) (15 min.)

(4) Plenum discussion on experiences, potentials & limitations (15 min.)